

The Value of Software[†]

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Software is one of the most important assets that needs to be priced in the digital economy. It has emerged as a disruptive technology, with companies primarily valued for their software offerings growing from 2 percent to 13 percent of market share between 1996 and 2023. We document persistent anomalies in growth forecasts and stock returns for software companies, indicating significant deviations from rational expectations over multiple decades. Our findings are consistent with Bayesian investors gradually learning about software’s growing importance, highlighting how markets can be very slow to discern fundamental shifts from transient shocks in noisy data. (JEL D22, G12, G32, L14, L86)

Software, the set of instructions and programs that tell computers what to do, has not only become integrally woven into every traditional industry over the past few decades but also has supported profound advances in the modern economy through digital innovations. Software has emerged as one of the most disruptive and important technologies in the digital age. Marc Andreessen penned the now infamous phrase “software is eating the world” and stated, “My own theory is that we are in the middle of a dramatic and broad technological and economic shift in which software companies are poised to take over large swathes of the economy,” (Andreessen 2011). This shift is evident from the disproportionate increase in the market share of software companies, which grew from 2 percent in 1996 to 13 percent in 2023, as shown in Figure 1. The importance of the software industry is expected to continue, as highlighted by Microsoft CEO Satya Nadella’s statement in the company’s 2022:III earnings call: “Going forward, digital technology will be the key input that powers the world’s economic output.”¹

Given the profound importance of software in the new economy, a first-order question is whether market participants are valuing this intangible asset correctly.

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¹ Microsoft Corporation. “2022 Q3 Earnings Conference Call.” Transcript: <https://view.officeapps.live.com/op/view.aspx?src=https://cdn-dynmedia-1.microsoft.com/is/content/microsoftcorp/TranscriptFY22Q3> (accessed June 2022)

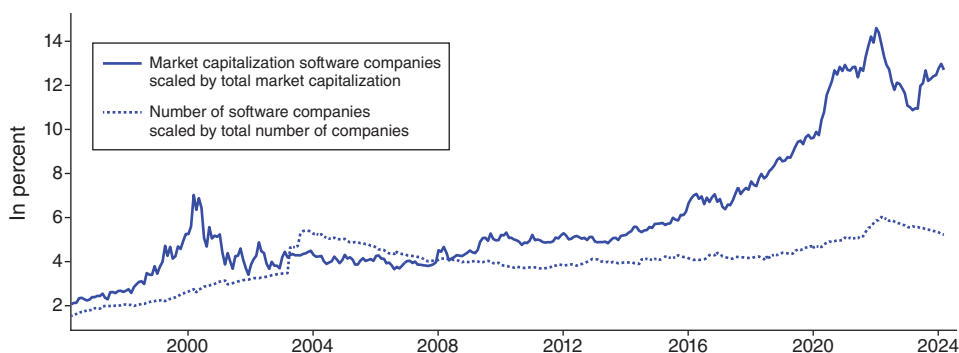


FIGURE 1. MARKET CAPITALIZATION OF SOFTWARE FIRMS

Notes: This figure shows the market capitalization of software firms scaled by the total New York Stock Exchange (NYSE), AMEX, and Nasdaq market capitalization. The dotted line charts the number of software firms scaled by the total number of market firms over the same period. The figure spans the period from January 1996 to December 2023.

This paper examines the valuation of software by market participants and the adequacy of their pricing models in ensuring efficient capital allocations. We find that over the past three decades, market participants have consistently undervalued software in a predictable manner. This is a staggering result as the efficient allocation of resources in a market economy relies on the accurate pricing of its securities (Hayek 1945), and software is arguably one of the most important and disruptive assets that needs to be priced.

Measuring the value of software presents challenges as transacted contracts for software offerings do not necessarily reflect the underlying value to the owner, and there is no established market for selling the overall rights to the software. However, by observing market prices that reflect the discounted future streams of software cash flows and forecasts about these cash flows, we can use standard asset pricing tools to determine whether the market is accurately valuing software. The main challenge in our analysis is linking software as a product to publicly traded companies that are valued primarily for their intangible software assets. We address this by focusing on companies where the majority of revenue streams come from software offerings, hereafter referred to as software companies.

To identify these software companies, we use highly granular revenue data for all US public companies from January 1996 to December 2023. From April 2003 onward, we utilize the FactSet RBICS Focus dataset, which analyzes each firm's annual revenue segment breakdowns, as reported in their 10-K filings, to determine the company's primary line of business. We then select those firms deriving at least 50 percent of their revenue from software licensing sales or subscriptions. For the period prior to April 2003, before the availability of the RBICS data, we review each firm's 10-K annual report to apply the same classification methodology. This selection process identifies 83 distinct software subindustries, encompassing a total of 1,207 US publicly listed software companies.²

²The firms that meet our software categorization criteria are primarily business-to-business software companies, such as Salesforce, Adobe, Datadog, and Microsoft. Our classification excludes consumer technology companies

We document persistent anomalies in growth forecasts and stock returns for software companies, indicating significant deviations from rational expectations over a prolonged period of nearly three decades. Both consensus and individual analyst forecasts consistently underestimate software companies' revenue growth rates. Specifically, realized growth rates are, on average, 2 percent higher than analysts' expectations across all forecast horizons, from the current fiscal quarter to three quarters ahead. This positive bias is not limited to outside equity analysts, as similar patterns are observed in the forecasts of company executives.

We also observe abnormally high stock returns for software companies during this same period. A simple buy-and-hold strategy investing in software companies, rebalanced monthly, yields annualized abnormal returns exceeding 4 percent, which remain largely unchanged after accounting for exposure to common pricing factors, such as those in the Fama-French three-factor (FF3) and Fama-French/Carhart four-factor (FF4) models. These abnormal returns are not driven by a few large firms; in fact, the estimates remain positive and are even higher when computing equally weighted (EW) portfolios instead of value-weighted (VW) portfolios. We do not observe similar anomaly patterns in other companies, such as nonsoftware growth firms or nonsoftware companies within closely related industries.

We supplement our main analysis with four additional tests. First, we document large, predictable positive drifts in software companies' stock prices around revenue announcements, persisting for several quarters. These drifts align with forecast errors, indicating that investors had not fully incorporated this information into prices. Second, we refine our classification of software companies by including firms with any software-derived revenue. We find a monotonic increase in forecast errors and abnormal returns as the proportion of software revenue rises, with the strongest effects in companies deriving 100 percent of revenue from software. Third, we find that nonsoftware companies utilizing software in their operations also experience positive forecast errors and abnormal returns, albeit smaller in magnitude, suggesting predictable spillover effects on software companies' principal customers. Lastly, we extend our sample back to the 1970s using an alternative classification method and find similar anomalies over this extended period.

The evidence that a single industry has consistently outperformed beliefs for three decades challenges the notion of full-information rational expectations (FIRE). Under FIRE, agents are assumed to fully understand and observe the model generating software growth rates, including parameters, states, and shocks. This full-information assumption eliminates the possibility of systematic anomalies, as expectations are determined by the true dynamics of the model and, on average, equal realizations. Reconciling our findings with FIRE would require arguing that expectations consistently turn out to be incorrect due to a 30-year streak of positive shocks occurring between expectation formation and actual outcomes. However, the probability of such prolonged, one-sided shocks is exceedingly small.

with a software or software platform component, such as Netflix, Uber, Apple, and Amazon, as their primary offerings are consumer goods or services, even if they are supported by software platforms. In supplementary analyses, we document predictable spillover effects on the principal customers of software companies.

We investigate alternative explanations for our findings that deviate from FIRE. The most plausible involves investors learning about the long-term characteristics of software growth rates, in line with the Bayesian learning literature (e.g., Farmer, Nakamura, and Steinsson 2024; Johannes, Lochstoer, and Mou 2016; Singleton 2021; Timmermann 1993; Li, Van Nieuwerburgh, and Renxuan 2023). This explanation posits that the software sector's performance was unexpected, leading to systematic forecast errors and excess returns. In 1996, market participants could not foresee the large-scale adoption of software technology by 2023; instead, they gradually learned about this trend from noisy data. To formalize this intuition, we develop and estimate a Bayesian learning model and show that agents' learning about software's growing importance can quantitatively account for the persistent anomalies we document over three decades.

Building on the long-run risk literature (Bansal and Yaron 2004; Bansal, Kiku, and Yaron 2016; Schorfheide, Song, and Yaron 2018), we model the growth rates for software and nonsoftware companies as being exposed to both a persistent common growth component, linked to the conditional expectation of consumption growth, and idiosyncratic growth shocks. We incorporate stochastic volatilities in both persistent and idiosyncratic components, which are important for matching asset prices and accurately tracking the data.

We assume that agents are rational but lack complete knowledge of the model's parameters and states. We position these agents at the start of our sample period in 1996, endowing them with plausible initial beliefs based on information available at that time. Specifically, we center the 1995:IV initial beliefs for software growth rates around parameter estimates of nonsoftware growth rates derived from 1961–1995 data. This reflects analysts' understanding of cash flow dynamics before the rise of software. As new data arrives, agents update their beliefs using Bayes' Law. Importantly, we allow agents to adjust their priors to the newly updated beliefs before each new observation. Priors thus evolve based on incoming information, remaining centered on the “right” location in the parameter space given the latest available information.

Our model of Bayesian agents learning about growth rates generates systematically biased forecast errors for software companies and unbiased estimates for nonsoftware companies, mirroring the data. To further strengthen the case for learning, we follow Farmer, Nakamura, and Steinsson (2024) and demonstrate that our model accounts for other well-known forecast anomalies present in our data. Specifically, both the empirical forecasts from consensus and individual analysts and our model-generated forecasts exhibit positive autocorrelation. They also deviate from one-for-one movements with actual realizations as shown by Mincer and Zarnowitz (1969) regressions and display underreaction to new information as shown by Coibion and Gorodnichenko (2015) regressions.

To analyze the asset pricing implications of belief revisions, we embed the cash flow process into a representative-agent endowment economy, assuming Epstein and Zin (1989) preferences. When solving for asset prices, we employ the anticipated utility approach proposed by Kreps (1998), which has become the standard method for handling parameter uncertainty in the literature (see, e.g., Cogley and Sargent 2008a, 2008b; Piazzesi and Schneider 2009; Johannes, Lochstoer, and Mou 2016; Chernov, Lochstoer, and Song 2025). The model generates positive *ex post* average excess

returns for both software and nonsoftware companies. Importantly, belief updates generate a wedge between *ex post* average excess returns and *ex ante* expected excess returns, resulting in abnormal excess returns for software companies. The model explains about two-thirds of the observed alpha for software companies.

Our Bayesian framework generates persistent anomalies spanning several decades due to two key factors. First, agents must filter persistent components from noisy data (Johannes, Lochstoer, and Mou 2016), where each new data point provides only limited information about long-run growth dynamics. Second, the updated beliefs in 2023 turn out to be markedly different from the initial beliefs in 1996; indeed, the 2023 posterior growth rate of software was unlikely according to the 1996 prior. Agents had to gradually learn about software's growing importance over many years. We find that the updated beliefs in each period, while highly probable given the priors at that time, consistently show a slight tilt toward higher long-term software growth rates, leading to significant changes in beliefs when accumulated over many decades.

It is important to highlight that, from the agents' perspective, software is correctly valued in each period as they update their beliefs using Bayes' rule. As a result, agents expect their forecasts and excess returns to be free of anomalies. However, if agents lack full information about the cash flow process and initially underestimate software growth rates, growth surprises will be positively biased and predictable. Consequently, econometricians analyzing the data *ex post* will conclude that forecasts and excess returns exhibit substantial anomalies.

We also discuss three alternative explanations that deviate from FIRE and have explained forecast anomalies in other contexts. First, informational frictions in sticky (Mankiw and Reis 2002) or noisy information (Sims 2003; Woodford 2001; Maćkowiak and Wiederholt 2009) models are implausible in our setting. Sticky information is unlikely for professionals with strong incentives and proficiency to process current data, while noisy information models imply no individual-level anomalies, inconsistent with our findings. Second, departures from Bayesian updating, such as in diagnostic expectations models (Bordalo et al. 2020), fail to explain our observed large positive biases and individual underreaction in software growth forecasts, instead predicting unbiased errors and individual overreaction. Assuming inefficient use of available information is also less suited for professionals who have strong financial incentives to utilize data optimally. Lastly, explanations based on reputational or other distorted analyst incentives struggle to account for the systematic anomaly differences between software and nonsoftware companies. They also cannot explain why investors haven't exploited these persistent biases, given the documented abnormal returns that align with forecast anomalies.

Our paper contributes to a growing literature on Bayesian learning that explains forecast and return anomalies through rational updating. We focus on learning about major technological change, while recent studies have applied this framework to other relevant contexts. Farmer, Nakamura, and Steinsson (2024) study learning about nominal interest rates and GDP growth, Johannes, Lochstoer, and Mou (2016) explore learning about consumption dynamics and its implications for the aggregate stock market, and Gómez-Cram, Kung, and Lustig (2025) investigate learning about the stance of fiscal policy and its impact on Treasury markets. Li, Van

Nieuwerburgh, and Renxuan (2023) study learning about both the parameters of house price growth and the underlying state of the housing market. The occurrence of anomalies in these learning frameworks is closely tied to the specific characteristics of the underlying data analyzed in each study.

These studies examine persistent variables that experienced gradual shifts over extended periods, culminating in substantial changes when viewed retrospectively. Crucially, these shifts diverge from historical patterns, such as major technological change, substantial increases in long-term rates from pre-1950 data, significant reduction in growth volatility after World War II, or the rise in deficits that reversed the fiscal consolidations of the 1990s. In such scenarios, learning is slow, often spanning multiple decades, as economic agents must gradually discern between fundamental shifts and transient shocks from incoming data. Conversely, for variables exhibiting regular, repetitive patterns (like revenues from mature industries) and when agents have access to extensive historical data, information about model parameters accumulates rapidly. In these cases, any anomalies that arise due to learning are likely to be short-lived.

Our paper also relates to two strands of literature on technological innovation. The first strand examines the development and diffusion of new technologies (Bloom et al. 2023), such as computers (Autor, Levy, and Murnane 2003), broadband (Akerman, Gaarder, and Mogstad 2015), robots (Acemoglu and Restrepo 2020), and artificial intelligence (AI) (Agrawal, Gans, and Goldfarb 2019; Webb 2020; Babina et al. 2024, 2025). The second strand of literature focuses on the relationship between technological innovation and stock prices (Jovanovic and MacDonald 1994; Laitner and Stolyarov 2003, 2004; Kogan et al. 2017). Past studies have explored the effects of technology adoption in the internet and railroad revolutions (Pástor and Veronesi 2009), the IT revolution (Ward 2020), and digitalization (Chen and Srinivasan 2024). We contribute to this literature by studying the diffusion of the software industry, arguably one of the most important sectors in the twenty-first century especially with its recent rise of AI. We further demonstrate that the relevant information affecting firm values does not remain confined to the software sector but diffuses sluggishly across economically linked firms in both quantities, such as revenues and stock prices in the spirit of Cohen and Frazzini (2008).

The present study also adds to the literature that underscores the growing importance of intangible assets and challenges market participants face in accurately valuing intangible assets. Our findings relate to Eisfeldt and Papanikolaou (2014) and Crouzet and Eberly (2018) who show that intangibles have become an increasingly important part of the productive capital stock of US firms. Moreover, they support previous research by Edmans (2011) demonstrating how market participants do not fully value intangibles. In addition, a large literature has documented the market's tendency to underreact to firms' investments in research and development (R&D) (Chan, Lakonishok, and Sougiannis 2001; Lev and Sougiannis 1996; Eberhart, Maxwell, and Siddique 2004; Hirshleifer, Hsu, and Li 2013; Cohen, Diether, and Malloy 2013). For example, Cohen, Diether, and Malloy (2013) show that successful R&D investments, which are key drivers of innovation, are largely ignored by market participants, resulting in significant abnormal returns for affected firms. We contribute to this literature by documenting substantial anomalies in forecasts and returns for software companies, where R&D constitutes one of the largest expenditures.

As software creates and houses data, it is the key mechanism that drives the production and value of digital data. In this regard, the study contributes to the recent and growing literature that speaks to the value of data (Abis and Veldkamp 2024; Farboodi and Veldkamp 2021; Farboodi et al. 2022; Veldkamp 2023)—these studies argue that the most valuable firms in the United States are valued primarily for their data and their ability to use this data to create market power (Eeckhout and Veldkamp 2022). While data and software are distinct concepts, in practice, digital data do not exist without software: It is nearly impossible to have one without the other. Because attempting to value data is inherently complex, an advantage of focusing on software is that its value and impact can be directly measured through companies that create and sell only software. In turn, we can identify these companies using granular revenue-based subindustry classifications. That said, a portion of software firm market values will likely also reflect the data the firms hold and are privy to, which often supports the firm's future strategic directions.

I. Measuring Software

Measuring software inherently involves incorporating a notion of value. We define the value of a software product as the present discounted value of the cash flows it generates. Therefore, if we have market prices and forecasts of these cash flow streams, we can utilize standard asset pricing tools to evaluate whether market participants value software correctly. To this end, our approach involves identifying publicly traded companies whose primary source of revenue is generated from software offerings. Although the valuation of these companies may also reflect other factors, such as the value of fixed assets, they are predominantly valued for their intangible software assets, with the majority of their cash flow streams directly attributable to their software products.

We classify a company as a software company if it derives at least 50 percent of its revenue from software offerings. Our sample includes all US publicly traded companies from January 1996 to December 2023. For the period from April 2003 to December 2023, we utilize the FactSet RBICS Focus (2023) dataset to identify software companies. The FactSet RBICS methodology involves analyzing each firm's annual revenue segment breakdowns, as reported in its 10-K filings, to determine the company's primary line of business, defined as the sector generating the majority of its revenue. Its sector classifications are highly granular, encompassing 1,704 distinct revenue-based subindustries. Among these, we identify 83 software subindustries. Table A.1 in the Supplemental Appendix provides a detailed list of these software subindustries.³

To extend our sample period prior to the availability of the RBICS data in 2003, we classify companies from January 1996 to March 2003 using the same methodology employed by FactSet's RBICS system. First, we scanned each firm's 10-K

³ We reviewed FactSet's software categorizations from 2003 and 2023 and found two minor classification errors. In 2016, FactSet temporarily misclassified Honeywell International Inc. as a software company for two months. Similarly, PayChex Inc. was briefly misclassified under software for one year in 2003 and one month in 2023. We corrected these errors by treating both companies as nonsoftware companies during these periods. These adjustments have no impact on any of the study's findings or inferences. It's worth noting that such errors are rare in FactSet's system; we identified only these two cases over a 20-year period.

annual report to identify those firms whose business descriptions include a software component. This process flagged over 1,100 firms' annual reports to have a software component. We then manually review each of these flagged 10-Ks to determine if the company derived more than 50 percent of its revenue from software licensing sales or subscriptions in a given year. By combining the RBICS data from April 2003 onward with our manually extended sample from January 1996 to March 2003, we identify a total of 1,207 publicly listed software companies spanning the period from 1996 to 2023.

We mitigate concerns of forward-looking bias in our company classification by using historical unrevised data. For each firm, we utilize the revenue information from the most recent publicly available annual filing and update the classification accordingly when new filings become publicly available. Importantly, we use the unrevised revenue value at each point in time to ensure that we are using the value that was known at that particular point in time.⁴

We also address concerns about survivorship and backfill biases. The dataset includes not only currently viable entities but also companies that have merged, filed for bankruptcy, gone private, or ceased to exist. Additionally, if a new company enters the dataset with, potentially, the full history (introducing data not previously available), FactSet records the time of the change. Therefore, we only use available information when making the industry classification at each point in time. Hence, we can track real-time changes in companies' classifications as the company's business evolves and have the industry classifications for those companies that are subsequently delisted.

Figure 1 plots the total market capitalization of software firms as a percentage of the US total market capitalization over the sample period, alongside the number of software firms as a percentage of all US public firms. From 1996 to 2023, the market capitalization of software companies increased significantly from 2 percent to 13 percent. Software companies have gained a notably larger share of the total market capitalization since 2006. While the late 1990s and the 2022 and 2023 monetary tightening period experienced significant declines in software market share, both were followed by recoveries, with the latter significantly recovering by the end of 2023. As of December 2023, software companies account for only 5 percent of the total number of US public firms but represent 13 percent of the total market capitalization, underlining their emergence as one of the most influential and disruptive industries.⁵

Table A.2 in the Supplemental Appendix lists the ten largest software companies by market capitalization for each year from 1996 to 2023 and presents the combined

⁴Companies can include up to three years of historical segment data in their annual filings. FactSet and our pre-2003 extended dataset captured each of these updated values while retaining the original records. As a result, we have access to a point-in-time history of each revenue segment as reported for the first time by the company in each filing. To prevent any forward-looking bias, we only use the original segment classifications and not the respective updated segment classifications in subsequent annual filings.

⁵To further shed light on the growing influence of software on business operations, we examine the frequency of software mentions in quarterly earnings calls of US public companies. These calls are generally regarded as providing insights into the most critical issues confronting these organizations (Bushee, Matsumoto, and Miller 2003; Matsumoto, Pronk, and Roelofsen 2011; Hassan et al. 2019). Figure A.1 in the Supplemental Appendix shows the percentage of earnings calls that included discussions of "software," "digital," or "cloud" related topics. The figure reveals that the frequency of such discussions between firm executives and analysts increased substantially during our sample.

market share of the top three, five, and ten companies. The market share of the top ten software companies relative to the entire market exhibits a steady upward trend, growing from 1.76 percent in 1996 to 10.38 percent in 2023.

Distinct Attributes of Software.—Software is a new and disruptive asset class with distinct attributes. Firstly, as intangible assets, software firms have unprecedented potential for scalability and growth (Haskel and Westlake 2017; Crouzet et al. 2022). Once created and operational, software products can be instantly deployed through cloud services to all users simultaneously (Govindarajan, Rajgopal, and Srivastava 2018). Consequently, software companies typically maintain low inventory levels and sidestep logistical and input-pricing concerns common to manufacturing-based industries, facilitating rapid adoption and high growth rates.⁶

Secondly, software companies exhibit high persistence in revenue growth rates. With the adoption of cloud computing, software companies have shifted from selling perpetual software licenses, which represent a one-time sale, to offering software as a service, which utilizes a subscription model. This shift has resulted in remarkably stable revenues, as clients are obliged to pay for the software on a monthly or annual basis. Furthermore, software companies enjoy significant greenfield opportunities compared to other industries, owing to the industry's high level of innovation, which enables firms to enhance and expand their software offerings every year. As a result, these companies often have high net revenue retention rates in excess of 110 percent, implying that clients from the previous year contribute at least an extra 10 percent of revenues each year before considering the effects of new clients.

Thirdly, software companies benefit from increasing returns to scale on their intangible investments, resulting in both high and persistent revenue growth rates and gross margins. These firms' primary expenses are usually associated with research and development costs related to software development. Since the marginal cost of providing an additional unit of software is minimal relative to the sales price, the potential for scalability of software companies enables them to achieve high and persistent gross margins (Covarrubias, Gutiérrez, and Philippon 2020).

II. Valuing Software

This section documents persistent anomalies in growth forecasts and stock returns for software companies.

A. Forecasting Bias

We compute forecast errors, defined as the realization minus the forecast, and find a systematic positive bias in both the consensus and individual forecasts. This

⁶By way of example, in Snowflake's (2022, pp. 10, 15) fiscal year 2022 10-K report, they note that their platform can be "deployed across multiple public clouds and regions" and that they have the ability "to elastically scale up and scale down," emphasizing the "ease of deployment, implementation, and use" of its product offering ([https://s26.q4cdn.com/463892824/files/doc_financials/2022/q4/Snowflake-Inc.-files-\(10-K\)-Basic-annual-filing,-for-period-end-31-Jan-22-\(SNOW-US\).pdf](https://s26.q4cdn.com/463892824/files/doc_financials/2022/q4/Snowflake-Inc.-files-(10-K)-Basic-annual-filing,-for-period-end-31-Jan-22-(SNOW-US).pdf) (accessed June 2022))

indicates that, over the last three decades, financial analysts and firm executives have consistently underestimated software revenue growth rates.

Analysts' Consensus.—We focus on quarter-over-quarter revenue forecasts and the corresponding forecast errors. To construct our dataset of analyst expectations, we extract analyst-level revenue forecasts from the IBES detail history file (LSEG IBES 1996–2023), spanning from the first quarter of 1996 to the third quarter of 2023. We include all forecasts issued within a 45-day window following a company's announcement of its total fiscal quarter revenue. In instances where an analyst provides multiple forecasts for the same firm and fiscal quarter within this 45-day period, we retain only the most recent forecast to ensure that our analysis incorporates the latest estimates from each analyst.

During this 45-day window, analysts typically provide forecasts for the current fiscal quarter, as well as for the subsequent three quarters. To derive the analysts' consensus for each forecasting horizon, we calculate the median of these individual forecasts. The consensus forecast error is then determined by subtracting the consensus forecast from the actual revenue realization for that period. Since the forecast errors are in dollar terms, we scale them with the actual revenue value realized in quarter $t - 1$, which is included in the forecasters' information sets.⁷

We winsorize outliers by removing forecast errors that exceed the 99.9th percentile in absolute value for each forecast horizon. In our benchmark analysis, we also exclude forecast errors associated with small firms, defined as those ranking in the first decile based on the previous fiscal year's June market equity, according to NYSE breakpoints. The rationale for excluding forecast errors from small firms is that these companies typically do not provide guidance and have limited analyst coverage. We present the summary statistics of forecast errors in Table A.3 in the Supplemental Appendix.⁸

Let $y_{t+h|t}^{(k)} \equiv F_t y_{t+h}^k$ denote the h -period ahead of consensus forecast, made at time t , for the future revenue value $y_{t+h}^{(k)}$ of company k . The forecast error for this period is defined as $e_{t+h|t}^{(k)} \equiv y_{t+h}^{(k)} - F_t y_{t+h}^{(k)}$. To compute the bias in these consensus forecasts, we estimate a pooled panel regression using the following equation:

$$(1) \quad e_{t+h|t}^{(k)} = a_h + u_{t+h}^{(k)},$$

where the coefficient a_h measures the average bias at a given horizon h and $u_{t+h}^{(k)}$ is the error term. If forecasts are unbiased, then they should equal realizations on average, implying $a_h = 0$.

Column 1 in Table 1 presents our estimates of a_h for the set of software companies, with t -statistics provided in square brackets. Standard errors are clustered at both the firm and year-quarter levels. The estimates show a positive bias in analyst

⁷We find that the choice between using the median or mean forecast as the measure of consensus has little impact on the results. Table A.4 in the Supplemental Appendix compares the average forecast errors for software companies using these two measures of consensus, revealing a high degree of similarity. Given this finding, we focus on the median consensus forecasts throughout the remainder of the paper.

⁸We conduct robustness checks using different winsorization thresholds, including tests without any winsorization (Tables A.6 and A.7 in the Supplemental Appendix). We also explore alternative inclusion criteria, such as requiring firms to have more than two analysts following them, instead of excluding small firms' forecast errors (Table A.8 in the Supplemental Appendix). The results remain consistent across these robustness tests.

TABLE 1—BIASES IN THE CONSENSUS FORECAST

	Software I (1)	All nonsoftware II (2)	I–II (3)	Nonsoftware growth III (4)	I–III (5)
Forecast horizon: one quarter					
Bias	2.21	1.54	0.67	2.41	−0.20
<i>t</i> -statistic	[6.88]	[7.15]	[1.92]	[6.07]	[−0.47]
Observations	13,417	212,216	225,633	26,329	39,746
Forecast horizon: two quarters					
Bias	1.78	0.03	1.75	−0.24	2.02
<i>t</i> -statistic	[2.69]	[0.08]	[2.48]	[−0.29]	[2.24]
Observations	12,893	202,813	215,706	25,375	38,268
Forecast horizon: three quarters					
Bias	2.21	−1.41	3.61	−4.64	6.84
<i>t</i> -statistic	[2.16]	[−2.36]	[3.55]	[−3.71]	[5.08]
Observations	12,243	189,954	202,197	23,994	36,237
Forecast horizon: four quarters					
Bias	2.94	−2.25	5.19	−8.09	11.03
<i>t</i> -statistic	[2.06]	[−2.98]	[3.78]	[−4.54]	[5.61]
Observations	11,604	178,155	189,759	22,637	34,241

Notes: Columns 1, 2, and 4 present regression results for the equation: $e_{t+h|t}^{(k)} = a_h + u_{t+h}^{(k)}$, where $e_{t+h|t}^{(k)}$ is the forecast error for company k at horizon h , a_h is the average bias at horizon h , and $u_{t+h}^{(k)}$ is the error term. The regressions are run separately for the set of software companies, all nonsoftware companies, and nonsoftware growth firms. In columns 3 and 5, we estimate the following regression: $e_{t+h|t}^{(k)} = a_h + b_h \cdot \mathbf{1}_{t,sw}^{(k)} + u_{t+h}^{(k)}$, where $\mathbf{1}_{t,sw}^{(k)}$ is an indicator variable set to one if company k is classified as a software company in period t , and zero otherwise. The coefficient b_h represents the difference in average bias between software companies and the comparison group. Column 3 uses all nonsoftware companies as the comparison group, while column 5 uses only nonsoftware growth firms. The *t*-statistics, calculated using standard errors clustered by both firm and date, are reported in square brackets. The forecast errors are expressed in percent. The sample spans from 1996:I to 2023:III.

consensus forecasts for software revenue growth rates. Consensus forecasts underestimate the true revenue growth rate by an average of 2 percent, with these biases being statistically significant at the 1 percent level.

Column 2 in Table 1 reports the estimates of a_h for nonsoftware companies (i.e., the rest of the firms traded on US exchanges). The mean forecast errors decrease monotonically from 1.54 (*t*-statistic = 7.15) at a one-quarter horizon to −2.25 (*t*-statistic = −2.98) at a four-quarter horizon, suggesting that at longer horizons, analysts' forecasts are overly optimistic. Furthermore, column 3 presents the difference in forecast errors between software and nonsoftware companies. Across all forecasting horizons, the differences in average bias between software and nonsoftware companies are positive and statistically significant, as evidenced by the *t*-statistics exceeding two.

In column 3 of Table 1, we use all nonsoftware companies as the comparison group for software companies. However, this comparison may not be entirely fair, given that the last twenty years have been exceptionally favorable for growth companies and particularly challenging for value companies. Consequently, the positive bias in forecast errors observed for software companies might simply reflect the well-known growth miracle that has occurred over the past two decades. To address

this potential issue, we proceed by comparing the forecast errors of software companies to those of nonsoftware growth companies.

We classify a company as a nonsoftware growth company if it belongs to the first decile of portfolios sorted on Book-to-Market and if it is not classified as a software company, meaning it generates less than 50 percent of its revenue from software. To construct the Book-to-Market portfolios, we follow Fama and French (1993) and calculate the Book-to-Market ratio at the end of each June using NYSE breakpoints, where the book value in June of the current year is the book equity for the last fiscal year-end, and the market value is determined by multiplying the price and shares outstanding at the end of December of the previous year.

Column 4 of Table 1 presents the average forecast error for nonsoftware growth firms. At the one quarter forecast horizon, the consensus forecast for nonsoftware growth companies exhibits a positive bias similar to that found for software companies. However, the forecasting bias for nonsoftware growth companies diverges from that of software companies as the horizon lengthens, becoming negative and large for horizons beyond two quarters. The contrast is particularly striking at the four-quarter horizon, where analysts overestimate revenue growth rates for nonsoftware growth companies by -8.09 percent (t -statistic $= -4.54$) while underestimating them for software companies by 2.94 percent (t -statistic $= 2.06$, see column 1). The differences in means between software and nonsoftware growth companies are reported in column 5. Beyond the second forecasting horizon, the difference is positive and statistically significant, with t -statistics above 2.

Figure 2 shows the average forecast bias for every quarter in our sample for software companies, nonsoftware growth companies, and all nonsoftware companies. Each line in the figure represents the average forecast error across the current quarter and the subsequent three quarters, while the NBER recession dates are indicated by gray shaded bars. Although our sample starts in 1996, given IBES coverage is fairly sparse prior to 1998, we start the forecast bias figures in 1998—the first year we have a considerable number of revenue forecasts.

Figure 2 reveals significant differences in the forecast errors between the three groups. After 2006, the forecast errors for software companies are consistently large and positive across all quarterly forecast horizons, in stark contrast to the errors for all nonsoftware companies and nonsoftware growth companies. The latter two groups generally fluctuate between -5 percent and 5 percent over the sample period, with a negative average. To better highlight the persistent positive forecast errors for software companies in the post-2006 period, Figure A.2 in the Supplemental Appendix zooms in on this specific timeframe.

All three company groups experienced abnormally negative forecast errors around recessions, consistent with findings by Gómez-Cram (2022), who shows that analysts tend to be overly optimistic at the onset of economic downturns. Notably, before the 2001:I–2001:IV recession, forecast errors are large and positive, then reverse to abnormally large negative errors that persist for a couple of years following the recession. This pattern was especially pronounced in software and nonsoftware growth companies, likely reflecting the impact of the dot-com boom and bust from 1997 to 2002. During the 2019:IV–2020:II recession, software companies exhibit positive biases, suggesting higher-than-expected reliance on software throughout the coronavirus pandemic, while the other two groups exhibit negative forecast errors.

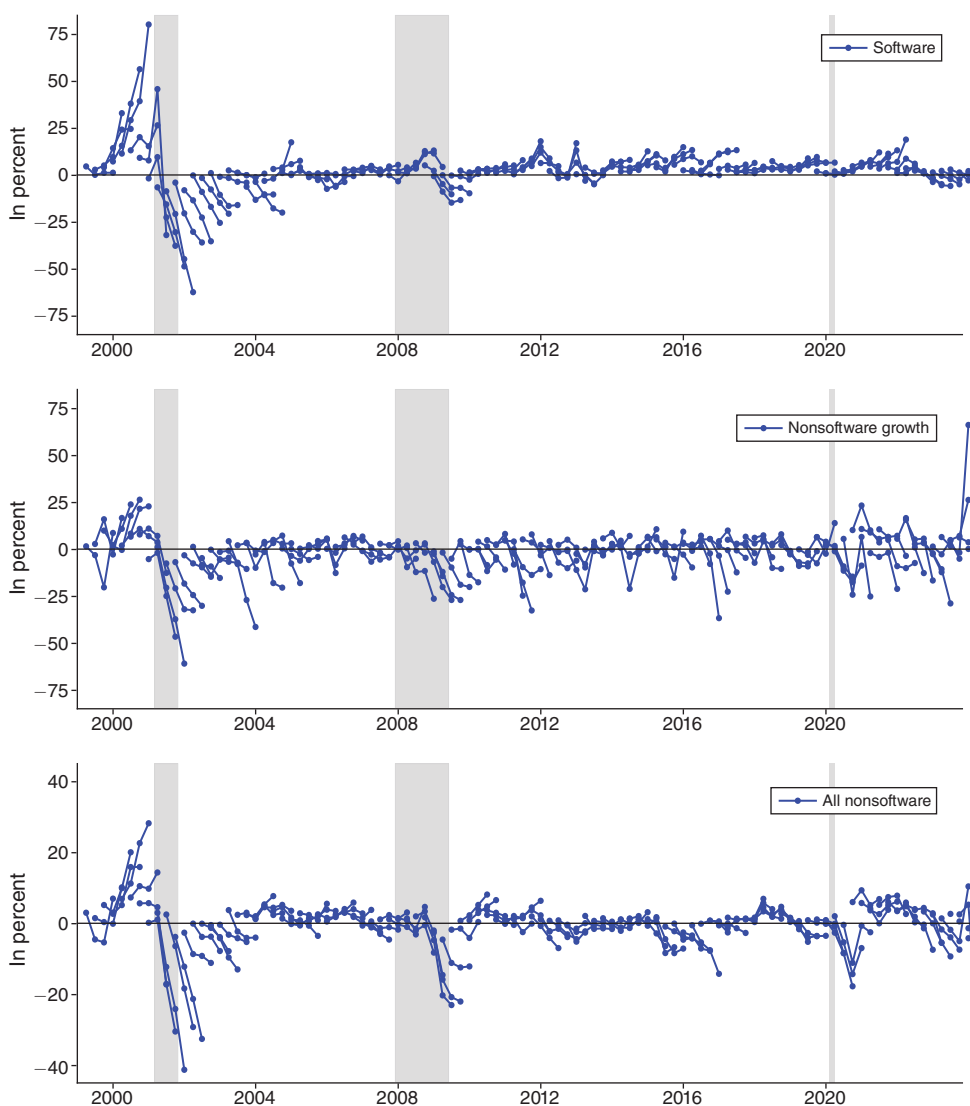


FIGURE 2. BIASES IN THE CONSENSUS FORECAST

Notes: This figure shows the average forecast bias for every quarter in our sample for software companies, nonsoftware growth companies, and all nonsoftware companies. Each line in the figure represents the average forecast error across the current quarter and the subsequent three quarters, while NBER recession dates are indicated by gray shaded bars. The figure spans the period from 1998 to 2023.

Table A.5 in the Supplemental Appendix presents average forecast errors for different subsamples. We divide the 28-year sample (1996 to 2023) into three subperiods: two 10-year periods (1996–2005 and 2006–2015) and one 8-year period (2016–2023). In line with Figure 2, we find that forecast errors for software companies are positive and large in the 2006–2015 and 2016–2023 subsamples. For instance, at the four-quarter horizon, forecast errors are 5.41 percent and 4.82 percent, respectively, with t -statistics above 2. For the pre-2006 period, average consensus forecast errors for software companies are insignificant, indicating that the large

swings in forecast errors observed during the dot-com boom and bust (as shown in Figure 2) largely offset each other.⁹

Individual Analyst Forecasts.—The previous section focused on the median consensus forecast. In this section, we present similar results for individual forecasters.¹⁰ We then show that there is learning within individual forecasters, as the extent to which they underestimate software revenue growth declines over time. However, despite this learning process, the forecasting bias for software companies remains positive and high even after eight years of producing quarterly forecasts for the same firm.

To analyze forecast error bias at the individual level, we compute individual forecast errors $e_{t+h|t}^{(k,i)} \equiv y_{t+h}^{(k)} - F_t y_{t+h}^{(k,i)}$, where $F_t y_{t+h}^{(k,i)}$ is the forecast of analyst i for firm k . We then pool forecasters and companies to estimate a common bias coefficient. Panel A of Table 2 presents the results with standard errors clustered by forecaster and year quarter.

Column 1 shows that the estimated bias for software companies is positive and statistically significant at the 1 percent level for all quarterly horizons, echoing our results in Table 1 using the analyst consensus forecasts. However, the estimated forecasting bias for all nonsoftware companies, shown in column 2, is only positive and statistically significant for the first quarter, with a smaller magnitude of 1.32 (t -statistic = 7.42) compared to the estimated coefficient for software companies. For longer horizon forecasts, the bias for all nonsoftware companies is close to zero and not statistically significant, suggesting that individual analysts' forecasts for nonsoftware companies are unbiased at these longer horizons. Column 4 presents similar results for nonsoftware growth companies. Like nonsoftware companies, nonsoftware growth companies exhibit positive biases for shorter horizons but have unbiased forecasts at longer horizons. Moreover, as shown in column 5, the differences in means between software and nonsoftware growth companies are positive and statistically different from zero across all quarterly horizons.

Panel B of Table 2 presents the median value from forecaster-by-forecaster average bias estimates, following the approach of Bordalo et al. (2020). For each analyst, we compute the average forecasting bias, yielding a distribution of individual biases. We then take the median bias as an indicator of whether the majority

⁹Splitting the sample into pre- and post-2006 periods is a natural choice, as 2006 signaled the dawn of the commercial cloud computing era, which transformed the way software offerings were deployed and implemented across various regions. Amazon officially launched its Amazon Web Services (AWS) in the summer of 2006, although some customers had been using AWS under nondisclosure agreements prior to this. The term “cloud computing” gained significant traction in 2011, and the years following 2012 witnessed rapid advancements in this field, further supporting the growth of the software industry. The CEO of Okta, during a March 28, 2022, conference call, highlighted the impact of cloud computing on software distribution, stating, “It just works. You turn it on. There’s no deployment with cloud SaaS (software as a service) companies. So, simplicity is a big, big differentiator.” (Wolfe Research Virtual “March Madness” Software Conference, March 28, 2022. <https://seekingalpha.com/article/4497426-okta-inc-s-okta-ceo-todd-mckinnon-on-wolfe-research-virtual-march-madness-software-conference>, accessed June 2022). Ewens, Nanda, and Rhodes-Kropf (2018) also use the introduction of cloud computing services by Amazon in 2006 as a technological shock that impacted companies with a strong online presence.

¹⁰We exclude forecasts where the forecast’s analyst ID variable (ANALYS) is set to zero. This exclusion is based on IBES’s claim in October 2018 that they had anonymized the identifications of their brokers and analysts in the detail history files by setting the forecast’s analyst ID variable to zero. However, this issue does not significantly impact our tests, as Law (2023) finds no evidence of extensive ID anonymization for the annual and quarterly forecasts of US firms, with no indication of a large-scale ID reshuffling for US company forecasts. Law (2023) finds that the anonymization primarily affects forecasts of non-US firms.

TABLE 2—BIASES IN INDIVIDUAL ANALYSTS' FORECASTS

	Software I (1)	All nonsoftware II (2)	I–II (3)	Nonsoftware growth III (4)	I–III (5)
<i>A. Pooled average bias estimates</i>					
Forecast horizon: one quarter					
Average bias	2.70	1.32	1.39	1.77	0.93
<i>t</i> -statistic	[13.69]	[7.42]	[6.98]	[8.81]	[4.67]
Observations	105,641	1,160,743	1,266,384	176,521	282,162
Forecast horizon: two quarters					
Average bias	2.52	0.35	2.17	0.92	1.60
<i>t</i> -statistic	[5.99]	[0.83]	[5.17]	[2.28]	[3.92]
Observations	76,775	860,577	937,352	129,413	206,188
Forecast horizon: three quarters					
Average bias	2.79	−0.01	2.80	0.14	2.65
<i>t</i> -statistic	[4.34]	[−0.02]	[4.42]	[0.29]	[4.55]
Observations	62,492	685,730	748,222	102,692	165,184
Forecast horizon: four quarters					
Average bias	3.18	0.11	3.07	−0.11	3.29
<i>t</i> -statistic	[3.98]	[0.15]	[3.84]	[−0.18]	[4.55]
Observations	53,162	545,812	598,974	81,712	134,874
<i>B. Median bias from forecaster-by-forecaster average bias estimates</i>					
Forecast horizon					
One quarter	1.88	1.01	0.87	1.25	0.63
Two quarters	1.46	0.23	1.23	0.59	0.87
Three quarters	0.98	−0.00	0.98	0.01	0.97
Four quarters	0.77	0.07	0.69	0.02	0.75

Notes: Columns 1, 2, and 4 present regression results for the equation: $e_{t+h|t}^{(k,i)} = a_h + u_{t+h}^{(k,i)}$, where $e_{t+h|t}^{(k,i)}$ denotes the forecast error of analyst i for firm k , a_h is the average bias at horizon h , and $u_{t+h}^{(k,i)}$ is the error term. The regressions are run separately for the set of software companies, all nonsoftware companies, and nonsoftware growth firms. In columns 3 and 5, we estimate the following regression: $e_{t+h|t}^{(k,i)} = a_h + b_h \cdot \mathbf{1}_{t,sw}^{(k)} + u_{t+h}^{(k,i)}$, where $\mathbf{1}_{t,sw}^{(k)}$ is an indicator variable set to one if company k is classified as a software company in period t , and zero otherwise. The coefficient b_h represents the difference in average bias between software companies and the comparison group. Column 3 uses all nonsoftware companies as the comparison group, while column 5 uses only nonsoftware growth firms. The *t*-statistics, calculated using standard errors clustered by both firm and date, are reported in square brackets. Panel B reports the median forecast bias from forecaster-by-forecaster regressions. In all panels, the forecast error is expressed in percent, and the sample spans from 1996:1 to 2023:III.

of forecasters over- or underpredict revenue growth rates, reporting results separately for each group of companies. We find that only software companies exhibit a consistently positive bias across all forecast horizons, indicating that the majority of analysts systematically underestimate software revenue growth rates. For the other company groups, the biases are smaller in magnitude and close to zero for forecast horizons above two quarters, suggesting that the individual analysts' forecasts for nonsoftware and nonsoftware growth companies are largely unbiased at longer horizons.

Examining the forecast errors made by individual analysts allows us to contrast the forecasting biases made by the same analysts on software companies against those on nonsoftware firms within related industry sectors, like “prepackaged software.” This comparison is possible because analysts typically specialize in

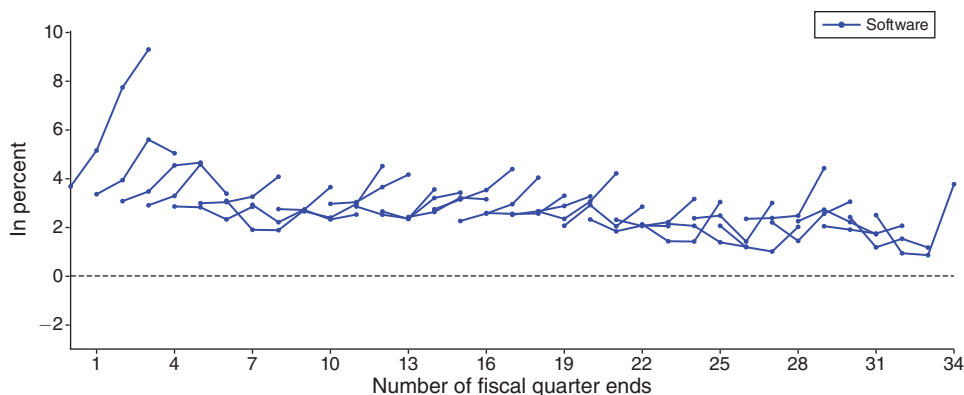


FIGURE 3. LEARNING WITHIN A SPECIFIC FORECASTER

Notes: This figure shows the average forecast error for software companies based on the number of prior forecasts an analyst i has issued for company k before time t . Each line represents the average forecast error for the current quarter and the subsequent three quarters. The sample covers the period from 1996:I to 2023:III.

specific sectors or groups of companies, rarely focusing on a single company.¹¹ Table A.9 in the Supplemental Appendix shows that, even in this stringent test, forecast errors remain positive for software companies and small for nonsoftware companies in similar industry sectors. For example, at the four-quarter horizon, the bias for software companies equals 3.22 percent with a t -statistic of 4.52, while the same analysts produce unbiased forecasts for nonsoftware companies, with the point estimate of the bias being just below zero and statistically insignificant ($a = -0.04$, t -statistic = -0.06).

Is there learning within a specific forecaster? To examine whether individual forecasters learn over time, we count the number of forecasts analyst i has made for company k before time t and compute the average forecast error for each level of experience. This analysis allows us to assess whether forecasting accuracy improves as analysts gain more experience with a specific firm. Figure 3 illustrates this relationship, with each line representing the average forecast error for the current quarter and the next three quarters. The slight downward trend in forecast errors over time suggests some learning. However, the errors remain consistently positive across all quarterly horizons, even after an analyst has been producing forecasts for the same firm for more than eight years. Table A.11 in the Supplemental Appendix provides the corresponding regression results.¹²

¹¹The top industries followed by analysts that issue forecasts about software companies in our sample are: prepackaged software (7,372), computer programming, data processing (7,370), semiconductors and related devices (3,674), computer integrated systems design (7,373), data processing and preparation (7,374), and computer facilities management (3,576), where the corresponding Standard Industrial Classification (SIC) codes are provided in parentheses. Table A.10 in the Supplemental Appendix shows that individual analysts' biases are close to zero and statistically insignificant for nonsoftware companies in these related industries. We obtain SIC industry codes from the US SEC (Security and Exchange Commission 2023) and supplement them with codes from Dun and Bradstreet (2023).

¹²We also investigate whether there is learning at the consensus level. Table A.12 in the Supplemental Appendix shows that consensus forecast errors for software companies decrease as the number of forecasting periods for each firm increases, indicating the presence of learning. However, the consensus bias remains positive even after over eight years of forecasts for the same firm.

TABLE 3—BIASES IN FIRMS' FORECASTS

	Software I (1)	All nonsoftware II (2)	I–II (3)	Nonsoftware growth III (4)	I–III (5)
<i>A. Pooled average bias estimates</i>					
Point estimate forecasts					
Average bias	2.56	1.19	1.37	1.65	0.91
<i>t</i> -statistic	[8.75]	[7.08]	[4.40]	[5.05]	[2.17]
Observations	9,173	47,274	56,447	7,954	17,127
Upper bound forecasts					
Average bias	0.72	−0.84	1.56	−0.27	0.99
<i>t</i> -statistic	[2.72]	[−4.80]	[5.62]	[−0.74]	[2.35]
Observations	7,976	37,447	45,423	6,120	14,096
<i>B. Median bias from firm-by-firm average bias estimates</i>					
Forecast					
Point estimate	2.12	1.07	1.05	1.17	0.94
Upper bound	0.81	−0.41	1.22	−0.09	0.90

Notes: Columns 1, 2, and 4 present regression results for the equation: $e_{t+h|t}^{(k)} = a_h + u_{t+h}^{(k)}$, where $e_{t+h|t}^{(k)}$ denotes the forecast error of firm k for its own performance, a_h is the average bias at horizon h , and $u_{t+h}^{(k)}$ is the error term. The regressions are run separately for the set of software companies, all nonsoftware companies, and nonsoftware growth firms. In columns 3 and 5, we estimate the following regression: $e_{t+h|t}^{(k)} = a_h + b_h \cdot \mathbf{1}_{t,sw}^{(k)} + u_{t+h}^{(k)}$, where $\mathbf{1}_{t,sw}^{(k)}$ is an indicator variable set to one if company k is classified as a software company in period t , and zero otherwise. The coefficient b_h represents the difference in average bias between software companies and the comparison group. Column 3 uses all nonsoftware companies as the comparison group, while column 5 uses only nonsoftware growth firms. The *t*-statistics, calculated using standard errors clustered by both firm and date, are reported in square brackets. Panel B reports the median forecast bias from firm-by-firm regressions. In both panels, the forecast error is expressed in percent, and the sample spans from 2001:I to 2023:III.

Figure A.3 in the Supplemental Appendix plots the same relationships for all nonsoftware companies and nonsoftware growth companies. Forecast errors for these two groups at longer horizons are initially negative but trend toward zero as the number of analyst forecasts increases. For nonsoftware companies, forecast errors even turn slightly positive, albeit remaining close to zero.

Firms' Forecasts.—The tests for forecasting biases can also be applied to firms' forecasts. Management-issued revenue forecasts could potentially be more accurate than those of analysts, as C-suite executives have a deeper understanding of their firm's business and performance.

To analyze firm executives' forecasts, we use the IBES Guidance database (LSEG IBES Guidance 2001–2023), focusing on the quarterly revenue guidance provided by company management. These forecasts are sourced from press releases and transcripts of corporate events, such as earnings calls and investor presentations. Managers typically issue forecasts as either point estimates or range estimates with a lower and upper bound. We compute forecast errors using the point estimates; when a range estimate is provided, we use the midpoint as the point estimate. We also calculate forecast errors using the upper bounds of range estimates. Due to data availability, the sample of revenue guidance spans from the first quarter of 2001 to the third quarter of 2023.

Panel A of Table 3 presents the average bias in forecast errors using both the point estimate forecasts and the upper bound forecasts. Standard errors are clustered

at both the firm and date levels. We start by describing the forecast errors using the point estimate forecasts. Column 2 shows that the management bias for nonsoftware companies is 1.19 percent (t -statistic = 7.08), while for software companies, as shown in column 3, it is 1.37 percentage points higher, with a t -statistic of 4.40 on the difference. Column 4 reveals that the bias for nonsoftware growth companies is only 1.65 percent (t -statistic = 5.05), which is 0.91 percentage points (t -statistic = 2.17) lower than the corresponding bias for software companies.

Positive firm forecast errors are often attributed to conservative management guidance, where management seeks to exceed market growth expectations by providing guidance that allows them to “beat and raise” analysts’ forecasts (Bartov, Givoly, and Hayn 2002; Skinner and Sloan 2002; Matsumoto 2002). To address this potential issue, we focus next on the upper bound of management’s forecasts, which represents the most optimistic forecast given their information set.

In contrast to the findings using the point estimate forecast, we observe that even the most optimistic guidance underestimates revenue growth for software firms by 0.72 percent per quarter (t -statistic = 2.72), as shown in column 1 of Table 3. However, for all nonsoftware companies, including nonsoftware growth companies, we find negative forecast biases, indicating that the most optimistic forecasts exceed the actual outcome.

Panel B of Table 3 presents the median value from firm-by-firm average bias estimates, following a similar approach as in Table 2 but using firms instead of individual analysts. The median bias indicates whether the majority of firms over- or underpredict their own revenue growth rates. Panel B shows that when using the mean forecast, the majority of firms underpredict their revenue growth, with software companies underpredicting by about 1 percentage point more than other firms. However, when considering the upper bound of management’s forecast, we find that the majority of software company managers continue to underpredict revenue growth rates, while the majority of managers at other companies tend to overpredict them.

Figure A.4 in the Supplemental Appendix shows the time series of one-quarter-ahead forecast errors in management guidance for software companies, all nonsoftware companies, and nonsoftware growth companies. The key takeaway is that software companies’ forecast errors have been consistently positive for two decades, while nonsoftware companies’ forecast errors are mostly centered around zero. These findings, along with those from above, suggest that software firm management systematically underestimates firm revenue growth rates, even when considering the most optimistic upper bound of their forecasts.

B. Stock Returns

This section shows that, over the past three decades, software companies have generated abnormally high stock returns relative to standard risk-factor models.

We obtain stock-level data from the Center for Research in Security Prices (CRSP) monthly stock file (Center for Research in Security Prices, 1996–2023b) and merge it with Compustat data (Standard and Poor’s 1996–2023) to obtain firm-level characteristics. The merge is performed using the CRSP-Compustat linking tables provided by Wharton Research Data Services (Center for Research in Security Prices 1996–2023c).

TABLE 4—PERFORMANCE EVALUATION AND ALPHAS

	Software I (1)	All nonsoftware II (2)	I–II (3)	Nonsoftware growth III (4)	I–III (5)
<i>A. Equally-weighted portfolios</i>					
Average return	18.54	10.93	7.61	7.05	11.48
<i>t</i> -statistic	[3.47]	[3.15]	[2.12]	[1.66]	[4.05]
Standard deviation	30.90	19.95	17.59	26.31	13.07
Three-factor alpha	5.71	−0.44	6.16	−5.59	11.30
<i>t</i> -statistic	[1.88]	[−0.34]	[2.38]	[−4.07]	[4.64]
Four-factor alpha	7.26	1.14	6.12	−3.33	10.59
<i>t</i> -statistic	[2.15]	[0.80]	[2.48]	[−1.91]	[4.78]
<i>B. Value-weighted portfolios</i>					
Average return	15.67	9.95	5.71	10.96	4.71
<i>t</i> -statistic	[3.35]	[3.32]	[2.11]	[2.91]	[2.49]
Standard deviation	26.04	15.86	17.76	17.68	15.78
Three-factor alpha	4.27	−0.58	4.85	0.81	3.46
<i>t</i> -statistic	[2.21]	[−1.74]	[2.37]	[1.23]	[1.78]
Four-factor alpha	4.29	−0.46	4.75	0.94	3.35
<i>t</i> -statistic	[2.18]	[−1.38]	[2.28]	[1.33]	[1.66]

Notes: This table presents performance evaluation measures and factor alphas for software companies, all nonsoftware companies, and nonsoftware growth companies. Panel A reports results for equally weighted portfolios, while panel B reports results for value-weighted portfolios. In both panels, column 3 reports results for a long-short portfolio that buys the software companies and sells all nonsoftware companies, while column 5 reports results for a long-short portfolio that buys software companies and sells nonsoftware growth companies. We annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1,200. *t*-statistics are reported in brackets. The table spans the period from January 1996 through December 2023.

We construct three distinct portfolios to investigate the stock market performance of software companies and provide a benchmark for comparison. The first portfolio is composed of software companies, the second consists of all nonsoftware companies, and the third comprises nonsoftware growth companies. We rebalance the portfolios monthly to account for changes in companies' classifications as their businesses evolve over time. To be included in any of the portfolios, a company must have a valid share price and number of shares at the formation date. Our analysis follows convention and employs closing prices to calculate returns from close to close and accounts for CRSP delisting returns. The sample spans from January 1996 to December 2023.

Column 1 in panel A of Table 4 presents standard performance measures for an EW portfolio of software companies. From January 1996 to December 2023, this portfolio achieved an average annual return of 18.54 percent with a standard deviation of 30.90 percent. In contrast, EW portfolios of nonsoftware companies (column 2) and nonsoftware growth companies (column 4) generated annualized mean returns of 10.93 percent and 7.05 percent, respectively. Columns 3 and 5 show that the differences in mean returns between software companies and the two other groups are statistically different from zero. The "long-short" portfolio, which involves buying software stocks and selling nonsoftware stocks, generated an average annualized return of 7.61 percent (*t*-statistic = 2.12), while the average annualized return from buying software and selling nonsoftware growth stocks equals 11.48 percent (*t*-statistic = 4.05).

Panel A of Table 4 reports factor alphas for EW portfolios, estimated from regressions of excess returns on the FF3 and FF4 models. We construct the risk

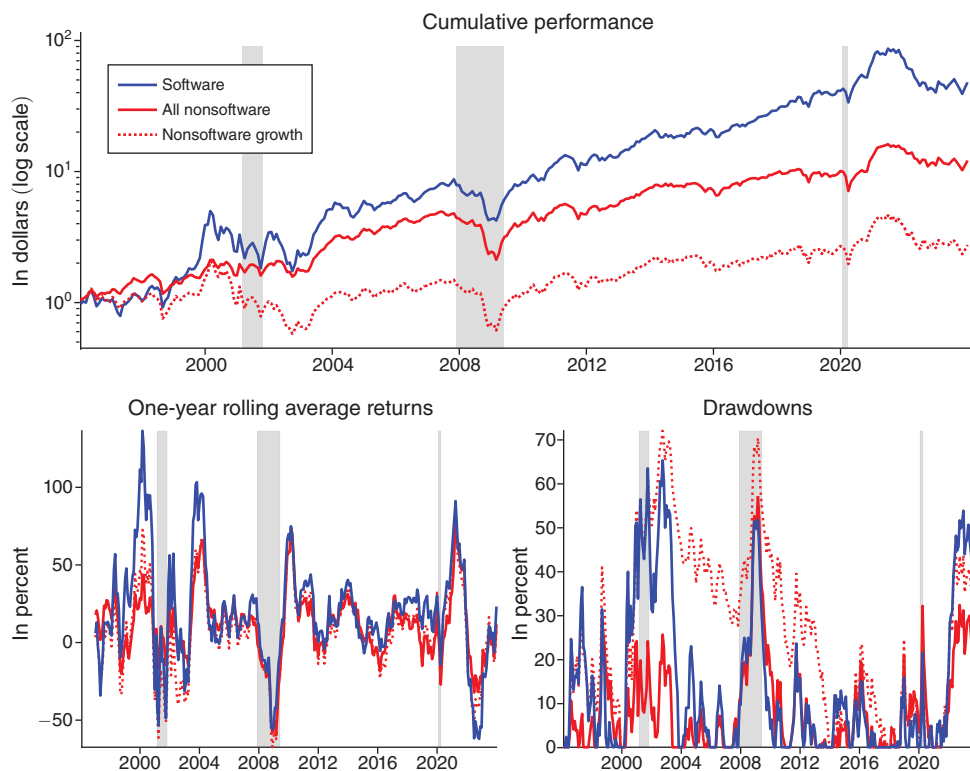


FIGURE 4. CUMULATIVE RETURNS ON EQUALLY WEIGHTED PORTFOLIOS

Notes: The top panel plots the cumulative returns to an equally weighted portfolio of software companies, nonssoftware companies, and nonssoftware growth companies from January 1996 to December 2023. The y-axis is on a log scale and all portfolios start with a \$1 investment in 1996, assuming no transaction costs. The lower-left panel plots rolling one-year returns for each portfolio, and the lower-right panel shows the drawdown of each portfolio, which measures the maximum loss from a peak to a trough before a new peak is attained.

factors following Fama and French (2023). The alpha represents the portion of returns that cannot be explained by exposure to common pricing factors. For the software portfolio (column 1) and the long-short portfolios (columns 3 and 5), the annualized FF3 and FF4 alphas are above 5 percent.

The top panel of Figure 4 shows the cumulative nominal returns from investing in EW portfolios of software, nonssoftware, and nonssoftware growth companies from January 1996 to December 2023. Assuming no transaction costs, the figure plots the cumulative returns to each strategy on a log scale, starting with a \$1 investment in 1996. The buy-and-hold strategy for software companies (dark blue line) significantly outperforms the other two portfolios over the sample period. By the end of December 2023, the software portfolio has grown to \$47.62, compared to \$12.06 for nonssoftware companies (red line), and \$2.70 for nonssoftware growth companies (red dotted line).

To shed light on the periods of under- and outperformance of software companies relative to the other portfolios, the lower panels of Figure 4 present the realized annual returns and maximum drawdowns for each portfolio. Drawdowns measure the maximum loss from a peak to a trough before a new peak is attained. The figure

shows that, after 2006, software companies have consistently generated larger annual returns than nonsoftware and nonsoftware growth companies. Moreover, except during the 2022–2023 tightening cycle, the drawdowns for the software portfolio are largely similar to those of the nonsoftware portfolio and smaller than those observed for nonsoftware growth companies. Figure 4 also highlights the impact of the dot-com boom and bust in 1997–2002, which led to large swings in annualized returns, with substantial positive returns during the boom followed by significant negative returns during the bust. As a result, software companies experienced considerable downturns during this period, requiring until the last quarter of 2003 to reach a new peak.

Panel A of Table A.13 in the Supplemental Appendix provides the subsample analysis. The mean returns and alphas for the long-short portfolios that buy software companies and sell nonsoftware companies are larger from 1996 to 2005, with annualized values of approximately 12 percent. However, these estimates are only significant at the 5 percent level for the four-factor alphas, suggesting that the large positive returns during the dot-com boom were not completely offset by the large negative returns during the bust, but these introduced substantial variance, making it difficult to obtain reliable estimates. For 2006 to 2023, the alphas are positive and statistically significant. Similarly, the long-short portfolio that buys software companies and sells nonsoftware growth companies yields positive and statistically significant mean returns and alphas across both subsample periods, with larger point estimates at the start of the sample.

The EW portfolio results are not driven by small and illiquid stocks. Specifically, Table A.14 in the Supplemental Appendix presents separately estimated alphas for microcaps (stocks below the twentieth percentile of NYSE market equity) and nonmicrocaps, following the definition in Hou, Xue, and Zhang (2020). We find significant alphas for both groups in the full sample, with particularly strong results in the 2006–2023 period.

We further demonstrate the robustness of the results using VW portfolios. Panel B of Table 4 presents the mean returns and alphas for VW portfolios of software, nonsoftware, and nonsoftware growth companies from January 1996 to December 2023. The factor alphas for software companies and the long-short portfolio that is long software and short all nonsoftware companies are positive (above 4.70 percent) and statistically significant (t -statistics above 2). Furthermore, the factor alphas for the long-short portfolio that sells all nonsoftware growth are positive (above 3.30 percent), although they are not statistically significant at conventional levels, with t -statistics below 1.80.

The subsample analysis presented in panel B of Table A.13 in the Supplemental Appendix shows that the VW portfolio of software companies outperformed the VW portfolio of nonsoftware growth companies in the pre-2006 sample. However, in the post-2006 sample, the mean return difference is positive but not statistically significant, reflecting the fact that the past twenty years have been exceptional for growth companies.¹³ These results suggest that the outperformance of growth

¹³ See also Figure A.5 in the Supplemental Appendix, which illustrates the cumulative returns, one-year rolling average returns, and the drawdowns of the VW portfolios. The annual returns of software companies consistently exceed those of nonsoftware growth companies before 2006. However, after 2006, the returns of these two portfolios closely track each other.

companies (relative to the market and value) post-2006 is concentrated in a couple of large companies. This is further supported by the finding that an EW portfolio of nonsoftware growth companies underperformed in the whole sample (column 3, panel A of Table 4) and in both the pre- and post-2006 sample periods (panel A in Supplemental Appendix Table A.13).

To further alleviate concerns about the growth miracle driving our results, Table A.15 in the Supplemental Appendix shows that software value companies outperform nonsoftware value companies. Moreover, Supplemental Appendix Table A.16 shows that the value-minus-growth returns among software firms are close to zero and statistically insignificant, suggesting that value and growth firms within the software sector performed similarly during this period. In contrast, the value-minus-growth returns among nonsoftware companies are negative and statistically significant since 2006, consistent with the growth miracle for nonsoftware companies observed during this period.

C. Additional Analyses

This section presents additional analyses and robustness tests, with detailed results and tables provided in Supplemental Appendix B. Each subsection is self-contained, enabling readers to selectively navigate through them without loss.

Stock Return Predictability around Revenue Announcements.—Section B.1 in the Supplemental Appendix documents that the stock prices of software companies exhibit large and predictable positive drifts around revenue announcements that persist for several quarters.

We document an asymmetric price reaction to negative and positive revenue surprises within minutes after the announcements. Specifically, we find that negative revenue news results in an initial large negative postannouncement return for software companies, larger than the one found for nonsoftware companies and nonsoftware growth companies. However, this negative price reaction for software companies is only temporary, as it is followed by significant positive abnormal returns in the subsequent four quarters, which completely reverse the initial price drop. In contrast, we find persistently positive returns over several quarters following positive revenue announcement news for software companies. Notably, for nonsoftware companies and nonsoftware growth companies, the price effects for both positive and negative announcements mostly pertain to the current revenue quarter.

The initial price dynamics we document are consistent with the findings in Lakonishok, Shleifer, and Vishny (1994) and Skinner and Sloan (2002), which show that growth firms experience an asymmetrically large negative price reaction to negative surprises. These papers argue that the large price decline reflects overly optimistic expectations, resulting in subsequently negative returns when those expectations are not met. However, an important difference in our findings is that the initial large negative price reaction for software firms reverses within two quarters, indicating that investors were initially too pessimistic and ended up overreacting to revenue surprises. This result also differentiates our findings from the large literature on postearnings announcement drift, as the price

drift for software firms persists for several quarters, whereas the postearnings announcement drift is typically concentrated in the quarter immediately following the earnings announcement.

Refining the Classification of Software Companies.—Section B.2 in the Supplemental Appendix refines the classification of software companies by considering all firms that report any source of revenue from software products, regardless of whether it constitutes the majority of their revenue. We find that the magnitude of both analysts' forecast errors and factor alphas increases monotonically with the percentage of revenue attributed to software, with the strongest effects observed for companies that derive 100 percent of their revenue from software offerings.

We sort firms into terciles based on their software revenue share, using segment data from FactSet RBICS with Revenue (2023), and examine the relationship between this measure and analysts' forecast errors and factor alphas. We find a monotonic relationship between analysts' forecast errors and the percentage of total revenue assigned to software by each company. For example, in Supplemental Appendix B.2, Table B.3 shows that firms in the highest tercile, which derive 100 percent of their income from software products, have an average forecast error of 4.26 percent (t -statistic = 3.13) at the fourth-quarter horizon, while firms in the lowest tercile, which derive around 10 percent of their revenue from software, have a close to 0 and statistically insignificant forecast error. Similarly, we find a monotonic relationship between the factor alphas and the revenue percentage attributed to software terciles. The annualized three-factor alpha for the highest tercile is 4.85 percent (t -statistic = 2.63), while the lowest tercile has an alpha close to 0 and statistically insignificant.

Customers of Software.—Section B.3 in the Supplemental Appendix investigates how market participants value nonsoftware companies that are the primary customers of software firms and, thus, benefit the most from software usage. We find that these companies also experienced positive forecast errors and alphas, albeit significantly smaller in magnitude compared to those found for software companies.

The rationale for focusing on the primary customers of software firms is that if the growth in software companies was a surprise year after year, then firms positively impacted by software should exhibit a similar growth surprise. This conjecture is grounded in previous research (Hong, Torous, and Valkanov 2007; Cohen and Frazzini 2008; Menzly and Ozbas 2010) showing that relevant information affecting firm values diffuses only gradually across structurally connected firms and industries. Hence, if market participants underreact to software, this information will also be sluggishly reflected across economically linked firms in both quantities, such as revenues, and stock prices.

Utilizing the FactSet Supply Chain Relationships (2023) database, which employs information from the Security and Exchange Commission (SEC) 10-K annual filings, investor presentations, and press releases to establish customer–supplier links between firms, we identify the primary customers of software companies. We find that customers with strong links to software companies exhibit positive forecast errors and factor alphas. While these values are smaller than those observed for software companies themselves, they are notably higher than the negative and

statistically significant values found for nonsoftware companies with no customer–supplier links to software. Supplemental Appendix Table B.4 presents the results.

Extending the Sample to the 1970s.—In Section B.4 in the Supplemental Appendix, we extend the sample to the 1970s. In our main analysis, we start our sample in 1996 due to the methodology used for classifying companies as software firms, which involves searching each company’s annual revenue segment breakdowns in their 10-K filings and identifying those that derive more than 50 percent of their revenues from developing and selling software. However, the SEC only mandated electronic filing of financial reports in the Electronic Data Gathering, Analysis, and Retrieval repository starting in 1995 or 1996, depending on the firm’s size. As a result, accurately classifying software companies using the revenue share approach is only feasible from 1996 onward.

Software, however, existed long before 1996. The adoption of computers began in the 1970s (Greenwood and Jovanovic 1999), utilizing various programming languages such as Unix, Fortran, Cobol, and FORMAC. In the 1980s, personal computers gained prominence, with Microsoft launching the Microsoft Disk Operating System in the early 1980s and the first version of Microsoft Windows in 1985. By the 1990s, software was widespread, although it had not yet achieved the immense scalability it would later experience with the advent of cloud computing.

To extend our analysis back to the 1970s, we employ an alternative classification method based on the SIC code 7372, prepackaged software, which is defined as “[e]stablishments primarily engaged in the design, development, and production of prepackaged computer software.” Companies self-select their SIC codes based on their primary line of business. In periods post-1996, where we have access to revenue segment breakdowns, approximately two-thirds of firms with the prepackaged software SIC code have more than 50 percent of their revenues coming from software. This suggests that the broad SIC code 7372 allows us to capture a significant share of software companies, although it is not a perfect classification.¹⁴

Using this alternative classification, we find that prepackaged software companies have generated abnormally high stock returns relative to standard risk-factor models over the past five decades, extending our findings from the main analysis. Supplemental Appendix Table B.6 shows the results for the extended 1976 to 2023 sample, with prepackaged software companies producing substantial annual returns and alphas that outperform non-prepackaged software companies and non-prepackaged growth companies. Furthermore, Supplemental Appendix Table B.7 provides a detailed breakdown of the returns for the prepackaged software portfolio from 1996 to 2023, showing that the superior performance is primarily attributable to those firms that derive the majority of their revenue from software activities.

¹⁴The SIC code 7372 is an imperfect classification of software companies. Around one-third of prepackaged software firms are misclassified, as their software revenue share is below 50 percent due to significant hardware, consulting, and/or service revenues, which make up the majority of their revenues. Moreover, 46 percent of our identified software firms come from other SIC industries.

III. Learning about Software

This section presents a model of Bayesian learning and demonstrates how analysts gradually learning about software's growing economic importance can quantitatively account for the persistent anomalies in both forecasts and returns documented earlier. Section IV discusses alternative explanations for these findings.

We assume that agents are rational but lack full information about the true model generating the data, following the Bayesian learning literature. We place our agents at the beginning of our sample period in 1996, endowing them with plausible initial beliefs about the parameters and states based on information available at that time. It is reasonable to assume that rational agents in 1996 did not anticipate the extensive adoption of software technology by 2023; instead, they had to learn about this trend gradually from noisy data. In essence, there was unexpected performance in the software sector; because it was unexpected, there were systematic forecast errors and excess returns.

A. Modeling Growth Rates

We model revenue growth rates for software and nonsoftware companies, $g_{s,t+1}$ and $g_{ns,t+1}$, respectively, as a joint process with levered exposure to a persistent predictable component x_t linked to the conditional expectation of consumption growth. We use consumption growth to lay the foundation for the subsequent asset pricing analysis in Section IIIC.

Specifically, the model we assume agents use to learn about and forecast growth rates is

$$\begin{aligned}
 (2) \quad g_{c,t+1} &= \mu_c + x_t + \sigma_{c,t} \eta_{c,t+1}, \quad \eta_{c,t+1} \sim \mathcal{N}(0, 1), \\
 x_{t+1} &= \rho x_t + \sqrt{1 - \rho^2} \varphi_x \sigma_{c,t} \eta_{x,t+1}, \quad \eta_{x,t+1} \sim \mathcal{N}(0, 1), \\
 g_{s,t+1} &= \mu_s + \phi_s x_t + \pi_s \sigma_{c,t} \eta_{c,t+1} + \sigma_{s,t} \epsilon_{s,t+1}, \quad \epsilon_{s,t+1} \sim \mathcal{N}(0, 1), \\
 g_{ns,t+1} &= \mu_{ns} + \phi_{ns} x_t + \pi_{ns} \sigma_{c,t} \eta_{c,t+1} + \sigma_{ns,t} \epsilon_{ns,t+1}, \quad \epsilon_{ns,t+1} \sim \mathcal{N}(0, 1).
 \end{aligned}$$

Following Bansal and Yaron (2004), the first equation decomposes consumption growth, $g_{c,t+1}$, into a persistent component, x_t , and a transitory component, $\sigma_{c,t} \eta_{c,t+1}$. The second equation defines the persistent component as an AR(1) process, where the parameter ρ determines the persistence of x_t . Both components, x_t and $\sigma_{c,t} \eta_{c,t+1}$, are not directly observable by analysts and need to be inferred from observed data.

The cash flow growth rates for software and nonsoftware companies have levered exposures to x_t through the parameter ϕ_i and to the transient idiosyncratic growth component, $\sigma_{c,t} \eta_{c,t+1}$, through the parameter π_i , as shown in the third and fourth lines of equation (2). We also allow for idiosyncratic movements in revenue streams as measured by $\sigma_{i,t} \epsilon_{i,t+1}$. The parameter μ_i denotes the unconditional mean of $g_{i,t+1}$.

We incorporate stochastic volatility in the dynamics of the persistent cash flow growth component, as well as in the independent shocks to consumption

and revenue growth rates. Specifically, we assume that the volatilities follow log Gaussian processes

$$\sigma_{i,t} = \varphi_i \sigma \exp(h_{i,t}),$$

where $h_{i,t}$ is a latent variable not directly observable by analysts that follows a linear autoregressive process

$$h_{i,t+1} = \rho_{h_i} h_{i,t} + \sigma_{h_i} w_{i,t+1}, \quad w_{i,t+1} \sim \mathcal{N}(0, 1) \quad \text{for } i \in \{c, s, ns\}.$$

The stochastic volatility process governing the persistent component plays an important role in determining asset prices, as highlighted by the long-run risks literature, while the other two volatility processes are important for tracking the dynamics of the revenue growth series.

To aid in identifying the stochastic volatility process of the persistent growth component, we utilize high-frequency monthly consumption data. However, recognizing the presence of measurement errors in this monthly data (e.g., Wilcox 1992; Slesnick 1998), we employ methods developed by Schorfheide, Song, and Yaron (2018) to sharpen inference about the persistent component.¹⁵ Finally, we use a mixed-frequency approach to link quarterly observed company growth rates with the monthly frequency of our model.

The resulting multivariate model of cash flow growth incorporates latent states, time-varying volatilities, mixed-frequency observations, and measurement errors. To estimate the model parameters and hidden states, agents employ Bayesian inference. The vector of parameters, denoted by Θ , is given by

$$\Theta = [\mu_c, \rho, \varphi_x, \sigma, \sigma_\epsilon, \sigma_\epsilon^q, \mu_s, \phi_s, \pi_s, \varphi_s, \mu_{ns}, \phi_{ns}, \pi_{ns}, \varphi_{ns}, \rho_{h_c}, \sigma_{h_c}, \rho_{h_s}, \sigma_{h_s}, \rho_{h_{ns}}, \sigma_{h_{ns}}]',$$

where σ_ϵ and σ_ϵ^q denote the standard deviation parameters related to the consumption measurement error model. The vector of latent states, $\mathcal{S}_t$, is represented by

$$\mathcal{S}_t = [x_t, h_{c,t}, h_{s,t}, h_{ns,t}]'.$$

To estimate the model, we specify a prior distribution for the parameters and latent states, which we discuss in the next section. Given this prior distribution, we use a Metropolis-within-Gibbs algorithm to sample from the posterior distribution of $(\Theta, \mathcal{S}_{0:T})$, where $\mathcal{S}_{0:T}$ denotes the sequence of latent states from time 0 to T . Supplemental Appendix C provides the data sources and a detailed description of the state-space representation and posterior inference.

¹⁵Nakamura, Sergeyev, and Steinsson (2017) provide complementary evidence for persistent variation in growth and volatility using annual data on a large panel of developed countries over a period exceeding 100 years.

B. Bayesian Learning and Initial Beliefs

We assume that agents do not know the values of the unobserved states $\mathcal{S}_t$ and parameters Θ . However, they possess initial beliefs about these quantities and update them using Bayes' law as new cash flow data become available.¹⁶

Initial beliefs form the starting point of learning. We specify initial beliefs in 1995:IV of the following form:

$$\begin{aligned} \mu_c &\sim \mathcal{N}(\tilde{\mu}_c, \tilde{\sigma}_{\mu_c}^2), & \rho &\sim \mathcal{N}^T(\tilde{\rho}, \tilde{\sigma}_{\rho}^2), & \varphi_x &\sim \mathcal{N}(\tilde{\varphi}_x, \tilde{\sigma}_{\varphi_x}^2), & \sigma &\sim \mathcal{IG}(\tilde{\alpha}_c, \tilde{\beta}_c), \\ \mu_i &\sim \mathcal{N}(\tilde{\mu}_i, \tilde{\sigma}_{\mu_i}^2), & \phi_i &\sim \mathcal{N}(\tilde{\phi}_i, \tilde{\sigma}_{\phi_i}^2), & \pi_i &\sim \mathcal{N}(\tilde{\pi}_i, \tilde{\sigma}_{\pi_i}^2), & \varphi_i &\sim \mathcal{N}(\tilde{\varphi}_i, \tilde{\sigma}_{\varphi_i}^2), \\ & & \rho &\sim \mathcal{IG}^h(\tilde{\alpha}_{h_c}, \tilde{\beta}_{h_c}), & \rho_{h_i} &\sim \mathcal{N}^T(\tilde{\rho}_{h_i}, \tilde{\sigma}_{\rho_{h_i}}^2), & \sigma_{h_i}^2 &\sim \mathcal{IG}(\tilde{\alpha}_{h_i}, \tilde{\beta}_{h_i}), \end{aligned}$$

for $i \in \{s, ns\}$ and $\mathcal{N}$, $\mathcal{N}^T$, and $\mathcal{IG}$ represent the normal, truncated (outside of the $(-1, 1)$ interval) normal, and inverse gamma distributions, respectively. Table C.1 in the Supplemental Appendix reports the initial beliefs along with the credible intervals.

These initial beliefs reflect analysts' prior knowledge and understanding of the cash flow dynamics as of 1995:IV, prior to the rise of software. We select priors that are reasonable given the information set available at that time. To establish these initial beliefs, we use data on nonsoftware companies and consumption from 1961:I to 1995:IV to estimate the cash flow model, assuming same growth rates for software and nonsoftware companies. We start in 1961:I because revenue data in the Compustat quarterly fundamentals files is only available from this year. Analysts then set their initial beliefs in 1995:IV equal to these parameter estimates, effectively centering their priors about software around the pre-1996 nonsoftware parameter estimates.

After setting their initial beliefs in 1995:IV, agents use the realized growth rates of 1996:I to update their beliefs. This period marks the first time agents have observed actual data for software companies. At the start of the next quarter, before observing the 1996:II data, we allow agents to update their initial beliefs by setting them equal to the newly computed posterior median estimates based on the 1961:I–1996:I sample. Thus, priors are dynamic and evolve based on incoming information, ensuring they remain centered on the “right” location in the parameter space given time t information. We continue this process quarter by quarter up to 2023:IV, generating a time series of prior and posterior distributions for both parameters and latent states.

Although the model operates at a monthly frequency and we have monthly consumption data, we update the parameters every three months to align with the quarterly frequency of the observed revenue growth rates. Furthermore, we exclude a few months of extreme observations around the COVID-19 pandemic

¹⁶ As in Farmer, Nakamura, and Steinsson (2024), we assume that agents know the model generating the data but not the values of the hidden states and parameters, which they learn sequentially using incoming data. Learning can be further complicated when agents must also learn about the model itself, such as the number of states in Markov switching models (Johannes, Lochstoer, and Mou 2016).

period and do not update the parameter and state estimates during this time as in Schorfheide and Song (2021). In the next two sections, we show that this Bayesian learning framework can account for the forecasting anomalies and alphas documented in Section II.

C. Forecasting Anomalies

Analysts utilize specification (2) to generate monthly revenue forecasts for software companies based on their current beliefs. The h -month forecast, conditional on information available at month t , is given by

$$(3) \quad \mathbb{E}_t g_{s,t+h} = \mu_s + \phi_s \rho^h x_t,$$

where μ_s represents the long-term forecast as $h \rightarrow \infty$, and $\phi_s \rho^h x_t$ captures short-term deviations from the long-term forecast, using the parameter estimates at time t . To assess whether the forecasts implied by equation (3) match the analysts' quarterly forecasting biases observed in the data, we use the posterior median estimates and aggregate the model-implied forecasts from a monthly to a quarterly frequency. The details are provided in Supplemental Appendix C.

Forecasting Bias.—This section shows that Bayesian agents learning about software and nonsoftware growth rates can generate systematic positive forecast errors for software companies and unconditionally unbiased estimates for nonsoftware companies, consistent with the data.

Figure 5 plots the average model-implied forecast bias for each quarter in our sample. Each line in the figure represents the forecast error for the current quarter and the subsequent three quarters, analogous to Figure 2 in Section IIA. The top panel displays the model-implied forecast errors for software companies, while the bottom panel shows the corresponding errors for nonsoftware companies. The model successfully captures the fact that analysts in the data consistently issue forecasts for software companies that are too low relative to realizations, with this bias being persistently positive after 2006, as in the data. In contrast, for nonsoftware companies, the forecast errors tend to be predominantly negative, especially for longer horizons. Notably, the model also captures the business cycle pattern in forecast errors, characterized by exceptionally large positive forecast errors preceding recessions, followed by exceptionally large negative forecast errors during recessionary periods.

Panel A of Table 5 compares the model-implied average forecast errors for software companies with estimates obtained using analyst consensus and individual analyst forecasts. The model successfully generates positive average forecast errors for software companies, closely matching the real-world estimated biases. For forecast horizons of two through four quarters, the model-implied biases are similar in magnitude to the “actual” forecast biases using analyst consensus or individual forecasts. However, for the one-quarter horizon, although the model-implied bias is positive and statistically significant, it is smaller than the bias in the data using analyst consensus or individual forecasts. The model-implied bias of 0.57 percent (t -statistic = 2.18) for the first quarter is more consistent

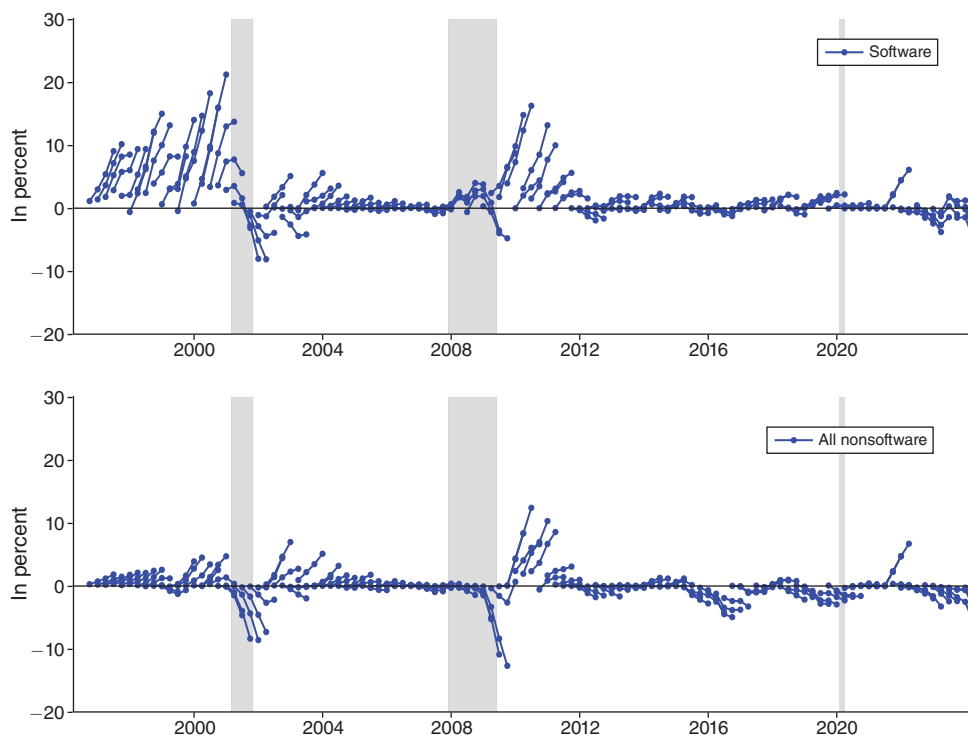


FIGURE 5. BIASES IN MODEL-IMPLIED FORECASTS

Notes: This figure shows the average model-implied forecast bias for every quarter in our sample for software companies, and all nonsoftware companies. Each line in the figure represents the model-implied average forecast error across simulations for the current quarter and the subsequent three quarters. NBER recession dates are indicated by gray shaded bars.

with the forecast errors made by firm managers using the upper-bound forecast, which equals 0.72 percent (t -statistic = 2.72; Table 3).

Panel A of Table 5 also presents the model-implied average forecast errors for nonsoftware companies. For forecast horizons of two through four quarters, the model-implied forecast errors closely resemble the individual forecasting biases computed in the data, as they are statistically insignificant and close to zero in both cases. However, the model is unable to account for the positive forecasting biases at the first-quarter horizon, which are significant in the data but statistically insignificant in the model.

Other Forecasting Anomalies.—This section examines whether analyst forecasts about software and nonsoftware growth rates in the data display other well-known forecast anomalies highlighted in the literature. We then evaluate our model's capability to replicate these anomalies. Although we know from Farmer, Nakamura, and Steinsson (2024) that various forecast anomalies about nominal interest rates and GDP growth can be successfully generated within a Bayesian learning framework, these tests still provide additional checks for the model's performance, as these

TABLE 5—FORECAST ANOMALIES: MODEL VERSUS DATA

	Software				Nonsoftware			
	Forecast horizon in quarters				Forecast horizon in quarters			
	1	2	3	4	1	2	3	4
<i>A. Bias</i>								
Model	0.57 [2.18]	1.33 [2.24]	2.11 [2.30]	2.91 [2.34]	−0.02 [−0.17]	−0.02 [−0.09]	0.01 [0.02]	0.04 [0.09]
Data consensus	2.21 [6.88]	1.78 [2.69]	2.21 [2.16]	2.94 [2.06]	1.54 [7.15]	0.03 [0.08]	−1.41 [−2.36]	−2.25 [−2.98]
Data individual	2.70 [13.69]	2.52 [5.99]	2.79 [4.34]	3.18 [3.98]	1.32 [7.42]	0.35 [0.83]	−0.01 [−0.02]	0.11 [0.15]
<i>B. Autocorrelation</i>								
Model	0.30 [2.87]	0.59 [6.53]	0.43 [2.85]	0.33 [1.60]	0.42 [4.79]	0.43 [5.17]	0.10 [0.80]	−0.13 [−1.15]
Data consensus	0.39 [3.09]	0.21 [4.10]	0.08 [3.10]	0.04 [1.36]	0.09 [6.58]	0.08 [5.56]	0.07 [4.60]	0.08 [4.76]
Data individual	0.32 [5.79]	0.23 [6.90]	0.17 [5.46]	0.12 [2.92]	0.26 [15.44]	0.21 [11.40]	0.18 [7.01]	0.13 [4.90]
<i>C. Mincer-Zarnowitz</i>								
Model	1.03 [0.11]	0.99 [−0.04]	0.93 [−0.27]	0.87 [−0.52]	0.96 [−0.84]	0.91 [−1.31]	0.83 [−1.88]	0.74 [−2.67]
Data consensus	0.98 [−0.86]	0.92 [−2.37]	0.91 [−2.68]	0.88 [−1.73]	0.86 [−16.14]	0.85 [−14.69]	0.83 [−13.36]	0.81 [−8.66]
Data individual	0.97 [−2.98]	0.92 [−3.77]	0.95 [−1.99]	1.01 [0.17]	0.83 [−16.52]	0.82 [−16.24]	0.82 [−15.66]	0.78 [−11.79]
<i>D. Coibion-Gorodnichenko</i>								
Model	−0.01 [−0.09]	0.10 [1.03]	0.15 [2.20]		0.11 [1.92]	0.26 [2.57]	0.40 [3.64]	
Data consensus	0.13 [8.37]	0.24 [7.65]	0.28 [6.22]		0.07 [6.07]	0.13 [5.23]	0.15 [6.04]	
Data individual	0.13 [6.62]	0.18 [4.38]	0.20 [4.09]		0.08 [5.95]	0.15 [4.00]	0.18 [4.58]	

Notes: This table presents four different tests of forecast rationality using model-implied forecasts, analyst consensus forecasts, and forecasts by individual analysts. The model-implied forecasts are derived using posterior median estimates. The t -statistics, presented in brackets, are relative to the following hypotheses: bias = 0, autocorrelation = 0, Mincer-Zarnowitz = 1, and Coibion-Gorodnichenko = 0. For the model-implied moments, Newey-West standard errors are used with the lag length selected as $L = \lceil 1.3 \times T^{1/2} \rceil$. For the data moments, standard errors are clustered by both firm and date. The forecast horizons are in quarters, and the forecast errors and forecast revisions used in the tests are expressed in percent. The sample spans from 1996:I to 2023:III.

moments do not enter the likelihood function of our state-space model and were not targeted when choosing the priors.

Under FIRE, forecast errors should be serially uncorrelated. We test this prediction in the data and in the model using the following regression:

$$(4) \quad e_{t+h|t} = a + be_{t|t-h} + u_{t+h}.$$

FIRE implies that forecast errors should be unbiased ($a = 0$) and unpredictable based on available information at time t ($b = 0$). Panel B in Table 5 presents the b estimates from regressions using actual forecast errors, computed with analyst

consensus and individual analyst forecasts, and model-implied forecast errors. In both the data and the model, forecast errors exhibit positive autocorrelation for horizons of one to three quarters, with lower autocorrelation at the four-quarter horizon, for both software and nonsoftware companies.

Under FIRE, forecasts should move one-for-one with actual realizations. We test this prediction in the data and the model using the following Mincer and Zarnowitz (1969) regression:

$$(5) \quad y_{t+h} = a + bF_t y_{t+h} + u_{t+h},$$

where $F_t y_{t+h}$ denotes the analysts' expectations about variable y_{t+h} . For the model regression, we use the model-implied forecasts, denoted by the conditional expectations operator $\mathbb{E}_t[\cdot]$. FIRE implies $a = 0$ and $b = 1$. Panel C of Table 5 presents the estimated b coefficients. In the data, the estimated b 's are just below one and decrease as the forecast horizon increases. In the model, the coefficient is slightly above one for the one-quarter horizon but falls below one for longer horizons, also decreasing as the forecast horizon increases.

Another test of FIRE, introduced by Coibion and Gorodnichenko (2015), involves regressing future forecast errors on current forecast revisions:

$$(6) \quad e_{t+h|t} = a + b(F_t y_{t+h} - F_{t-1} y_{t+h}) + u_{t+h}.$$

Under FIRE, forecast errors should be unpredictable using information available at the time the forecast is made, implying $b = 0$. However, if b is positive and significant, it indicates that the forecast is underreacting to new information. In this case, when forecast revisions are positive (negative), the forecast does not increase (decrease) enough, leading to future positive (negative) forecast errors. Conversely, if b is negative and significant, it suggests that the forecast is overreacting to new information. We apply these Coibion-Gorodnichenko regressions to both the consensus forecast and the forecasts from individual analysts. We then compare these estimates with the model-implied estimates.

Panel D of Table 5 presents the results of the Coibion-Gorodnichenko regressions. In the data, the coefficients are positive and statistically significant for both software and nonsoftware companies, indicating underreaction in individual and consensus analysts' forecasts. The model-implied forecasts also exhibit underreaction. For nonsoftware companies, the estimated coefficients are positive across all horizons, whereas for software companies, the estimated effects are statistically significant for longer horizons. At shorter horizons, the estimated effects are not statistically significant. Notably, the model successfully captures the pattern across forecast horizons, where the estimated coefficients increase with the forecast horizon, mirroring the pattern observed in the data.

Overall, the results in this section show that Bayesian agents learning about revenue growth rates can generate positive forecasting biases for software companies and negative forecasting biases for nonsoftware companies, while also accounting for other prominent forecast anomalies.

D. Abnormal Returns

We now embed the cash flow process (2) into an endowment economy to quantify the asset pricing implications of our learning framework. Specifically, we compute the returns and alphas of software and nonsoftware companies implied by our model and compare these values with those observed in the data.

Representative Agent's Optimization and Model Solution.—We consider a representative agent with Epstein and Zin (1989) recursive preferences, given by

$$(7) \quad U_t = \left\{ (1 - \beta) C_t^{1-1/\psi} + \beta \left(\mathbb{E}_t[U_{t+1}^{1-\gamma}] \right)^{\frac{1-1/\psi}{1-\gamma}} \right\}^{\frac{1}{1-1/\psi}},$$

where C_t is consumption at time t , β is the time discount factor, γ is the coefficient of relative risk aversion, and ψ is the elasticity of intertemporal substitution. The agent maximizes lifetime utility subject to the budget constraint $W_{t+1} = (W_t - C_t)R_{c,t+1}$, where W_t is the agent's wealth at time t and $R_{c,t+1}$ is the return on an asset that pays aggregate consumption as dividends each period. The endowment stream is governed by the cash flow process described in equation (2).

The Euler equation for any asset with gross real return $R_{i,t+1}$ is given by

$$(8) \quad \mathbb{E}_t[M_{t+1}R_{i,t+1}] = 1,$$

where $M_{t+1} = \beta^\theta (C_{t+1}/C_t)^{-\frac{\theta}{\psi}} R_{c,t+1}^{\theta-1}$ is the real stochastic discount factor, and $\theta = (1 - \gamma)/(1 - 1/\psi)$. To price the nominal payoffs of software and nonsoftware companies, which are modeled as leveraged consumption claims in equation (2), we use the nominal discount factor $M_{t+1}e^{-\pi_{t+1}}$, where π_{t+1} is the logarithm of the inflation rate. For simplicity, we assume a constant inflation rate.¹⁷

When solving for asset prices, we employ the anticipated utility approach proposed by Kreps (1998). In this framework, agents treat parameters as random variables during the learning process but consider them constants when pricing assets. While agents maximize expected utility, they are not fully rational. They act as if their parameter beliefs will remain unchanged indefinitely, despite historical evidence showing that these beliefs have evolved significantly over time as new data became available.

Fully Bayesian agents would treat the parameter vector Θ as a random variable for both learning and decision-making. Collin-Dufresne, Johannes, and Lochstoer (2016) show that priced parameter uncertainty can have significant and persistent asset pricing effects, as permanent shocks to beliefs constitute a source of subjective long-run risk. However, solving the complete pricing problem with priced

¹⁷ While more complex inflation dynamics, such as nonneutral effects on future expected growth (Piazzesi and Schneider 2007; Bansal and Shaliastovich 2013; Gómez-Cram and Yaron 2021) or variations in the cyclical properties of inflation (David and Veronesi 2013; Song 2017; Campbell, Pflueger, and Viceira 2020), are important for matching key bond market features and capturing moments at the interplay between stocks and bonds, like time-varying bond-stock correlations, they are not relevant for explaining the returns and alphas of software and nonsoftware companies.

TABLE 6—PERFORMANCE EVALUATION AND ALPHAS: MODEL VERSUS DATA

Name	Software			Nonsoftware		
	Data			Data		
	EW	VW	Model	EW	VW	Model
$E_T[r_{i,t+1}]$	18.54	15.67	11.77	10.93	9.95	8.53
$E_T[r_{i,t+1} - r_{f,t}]$	16.44	13.57	10.07	8.83	7.86	6.83
$E_T[\mathbb{E}_t[r_{i,t+1} - r_{f,t}]]$	10.73	9.31	6.19	9.27	8.44	6.23
α	5.71	4.27	3.88	-0.44	-0.58	0.60

Notes: This table reports asset pricing moments from the data and as implied by the model for both software and nonsoftware companies. We report average ex post returns, $E_T[r_{i,t+1}]$, average ex post excess returns, $E_T[r_{i,t+1} - r_{f,t}]$, average ex ante excess returns, $E_T[\mathbb{E}_t[r_{i,t+1} - r_{f,t}]]$, and alphas, α , where E_T denotes the sample average. For the data moments, we report these values using both EW and VW portfolios. To compute the average ex ante excess returns and alphas, we employ the FF3 model. For the model-implied moments, we use the posterior median estimates and fix the preference parameters to $\gamma = 10$, $\beta = 0.999$, and $\psi = 1.5$. In both the model and the data, the frequency is monthly, and we report annualized values.

parameter and state uncertainty in our context is computationally infeasible due to the curse of dimensionality. While priced parameter uncertainty would generate higher ex ante risk premia, it would not necessarily help in matching alphas, defined as the difference between ex post average excess returns and ex ante expected excess returns. As Johannes, Lochstoer, and Mou (2016) point out, alphas are primarily driven by the specific sequence of growth shocks realized during the sample period.

Using the anticipated utility framework, we derive time t asset prices by conditioning on the posterior median beliefs at time t . We solve the asset pricing equation in (8) using the Campbell and Shiller (1988) log-linear approximation for returns and the Schorfheide, Song, and Yaron (2018) log-linear approximation of the log Gaussian process $\sigma_{i,t} = \varphi_i \sigma \exp(h_{i,t})$. This approach allows for quick computation of the solution for each time period due to its reliance on analytical approximations. Lastly, we do not calibrate preference parameters to match specific asset-pricing moments; instead, we employ parameter values commonly used in the literature (see, e.g., Bansal and Yaron 2004; Johannes, Lochstoer, and Mou 2016): $\gamma = 10$, $\psi = 1.5$, and $\beta = 0.999$. To keep the main text concise and focused on the results, we provide the description of the model solutions in Supplemental Appendix C.

Results.—Table 6 reports asset pricing moments from the observed data and those generated by our model over the same period, ranging from January 1996 to December 2023. We present results for both software and nonsoftware companies.

The model generates positive ex post average excess returns, denoted by $E_T[r_{i,t+1} - r_{f,t}]$, where $E_T[\cdot]$ represents the sample average. For software companies, the realized average excess returns equal 10.07 percent in the model, compared to 16.44 percent and 13.57 percent in the data using EW and VW portfolios, respectively. For nonsoftware companies, the model generates realized average excess returns of 6.83 percent, closely aligning with the data values of 8.83 percent and 7.86 percent for EW and VW portfolios, respectively.

To understand the impact of parameter learning on alphas, it is useful to compute the sample average of the ex ante expected excess returns, denoted by $E_T[\mathbb{E}_t[r_{i,t+1} - r_{f,t}]]$. In the model, we can calculate this from the conditional covariance term of the SDF with respect to stock returns: $\mathbb{E}_t[r_{i,t+1} - r_{f,t}] = -\text{cov}_t[m_{t+1}, r_{i,t+1}] - \frac{1}{2}\text{var}_t[r_{i,t+1}]$, which is a function of the time t parameters. In the data, we use the FF3 model to estimate a measure of the ex ante expected excess returns. We define the alpha, α , as the difference between the ex post average excess returns and the ex ante expected excess returns:

$$(9) \quad \alpha = E_T[r_{i,t+1} - r_{f,t}] - E_T[\mathbb{E}_t[r_{i,t+1} - r_{f,t}]].$$

Under FIRE, $\alpha = 0$, as $r_{i,t+1} - r_{f,t} - \mathbb{E}_t[r_{i,t+1} - r_{f,t}]$ is a function of cash flow and volatility shocks with mean 0. However, parameter learning can generate a significant wedge between ex post and ex ante excess returns leading to nonzero alphas.

Table 6 shows that our model generates a 3.28 percent spread in alphas between software and nonsoftware companies. In the data, that same spread is 4.85 percent (VW). Hence the model explains about two-thirds of the observed alpha. In contrast, the alpha for nonsoftware companies is close to zero in the data and in the model.

Overall, positive surprises in belief updates for software companies lead to ex post average excess returns that are higher than the ex ante expected excess returns. In contrast, for nonsoftware companies, the variation in belief updates is not significant enough to generate a meaningful difference between ex post and ex ante excess returns. The next section analyzes the changes in belief updates for software and nonsoftware companies that give rise to persistent anomalies in forecasts and returns.

E. Initial and Updated Beliefs

Figure 6 illustrates the evolution of posterior median estimates for the unconditional mean growth rates of software (μ_s) and nonsoftware (μ_{ns}) companies, accompanied by 98 percent confidence intervals. The evolution of other parameters in Θ is presented in Supplemental Appendix Figure C.2.

At the beginning of the sample in 1996, agents hold identical beliefs for both software and nonsoftware companies. However, as new data become available, agents consistently revise their estimates of μ_s upward, as evidenced by the marked positive trend shown in the figure. In contrast, estimates of μ_{ns} exhibit only a minor positive trend during the sample period. Notably, it takes approximately eight years of data for agents to statistically differentiate between the unconditional mean estimates μ_s and μ_{ns} . Agents initially attribute the higher realized software growth rates to greater stochastic volatility rather than a higher unconditional mean μ_s , as shown in Supplemental Appendix Figure C.3

The substantial increase in the long-run growth rate of software companies, as shown in Figure 6, is consistent with agents learning about the growing importance of software in the economy, which ex post emerged as one of the most disruptive and significant technologies. The figure also demonstrates that the exceptional performance of the software sector was not anticipated at the beginning of our sample

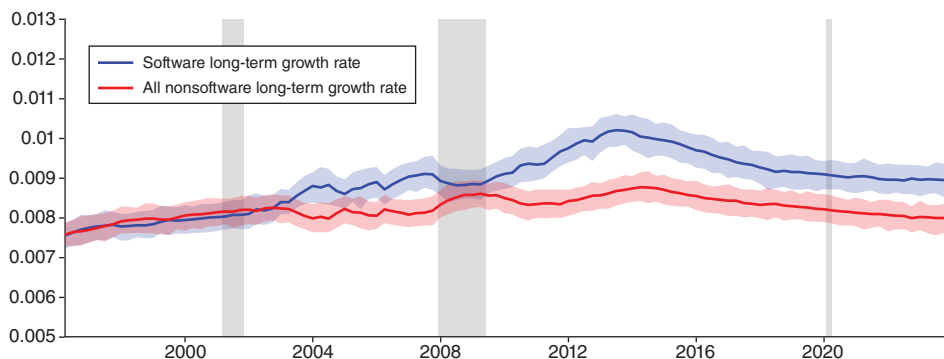


FIGURE 6. EVOLUTION OF THE LONG-RUN GROWTH RATE ESTIMATES

Notes: The figure shows the evolution of the estimated long-run growth rates for software, μ_s , and nonsoftware, μ_{ns} , companies. The straight line denotes the posterior median estimates and the shaded areas denote 98 percent confidence intervals. The figure spans the period from January 1996 to December 2023.

in 1996. The p -value of the final growth rate mean in 2023, given the initial beliefs established using pre-1996 data, is zero.

Therefore, agents in 1996 did not foresee the large-scale adoption of software technology by 2023; instead, they had to gradually learn about it over time. Table 7 reports the one-period p -values of the growth rate mean in $t + 1$, given the priors at time t . The priors evolve over time based on incoming information, centered around the new posterior median estimates. Columns 1 through 6 provide five-year snapshots of these one-period p -values, while the last three columns report their minimum, mean, and maximum values. From the agents' perspective, the time $t + 1$ posterior median estimates for μ_s are highly probable given their time t priors. After all, these estimates differ only by one period of noisy data. The average one-period p -value equals 0.47, indicating a slight tilt toward higher μ_s values relative to the prior distribution.

While each period's outcomes appear likely given the prior from the previous period, the total effect of these small, skewed parameter shifts leads to significant cumulative changes over time. Consequently, a prior deemed "reasonable" ex ante using pre-1996 information assigns a low probability to events occurring three decades later, especially given the major technological changes unfolding during this period.

It is important to highlight that, from the agents' perspective, software has been correctly valued in each period, as they employ Bayes' rule to update their beliefs with new information. As a result, agents expect zero-mean cash flow shocks and anticipate that their cash flow forecasts and stock returns will be free of anomalies at each point in time. However, if agents do not have full information about the cash flow process and initially underestimate the true long-run growth rates for software companies, then the growth surprises will be positively biased and strongly predictable by econometricians analyzing the data ex post. This would eventually lead econometricians to conclude that forecasts and excess returns exhibit substantial anomalies, even when carefully using only contemporaneous information in their analyses.¹⁸

¹⁸To see this, let $\mathcal{H}_t = \{g_t, g_{t-1}, \dots\}$ denote the available information up to time t . In each period, agents have prior beliefs $\mu_s \sim \mathcal{N}(\hat{\mu}_{s,t}, \sigma_{\mu_{st}}^2)$, with $\hat{\mu}_{s,t} = \mathbb{E}[\mu_s | \mathcal{H}_{t-1}]$ and $\sigma_{\mu_{st}}^2 = \text{var}[\mu_s | \mathcal{H}_{t-1}]$. Before observing

TABLE 7—ONE-PERIOD p -VALUES OF UPDATED BELIEFS GIVEN PRIORS

Parameter	Five-year snapshots						Min	Mean	Max
	1996	2000	2005	2015	2020	2023			
μ_s	0.35	0.46	0.24	0.57	0.55	0.47	0.16	0.47	0.86
μ_{ns}	0.31	0.45	0.70	0.62	0.57	0.52	0.16	0.49	0.71

Notes: This table presents one-period p -values of the growth rate mean in $t + 1$, given the priors at time t . To compute these p -values, we simulated draws from the time t prior distributions of μ_{ns} and μ_s , respectively, and calculated the percentage of draws that exceed the posterior median at time $t + 1$. Columns 1 through 6 provide five-year snapshots of these one-period p -values. The last three columns report the minimum, mean, and maximum values of these p -values over the entire sample period.

Learning can span several decades, as each new data point provides limited information about the long-run dynamics of growth rates. Over time, however, these incremental updates gradually shift beliefs closer to the true parameters. If these true parameters are constant, agents would eventually learn them fully, and the ex post documented anomalies would disappear. However, if these parameters are time-varying, as would be the case with evolving technological innovation within the software industry, then agents would constantly need to update their beliefs about these parameters.¹⁹

F. Alternative Initial Beliefs

Section C.7 in the Supplemental Appendix explores three alternative assumptions regarding agents' initial beliefs in our model. The first two scenarios examine agents with progressively more dispersed initial beliefs about Θ , while the third investigates a perfect foresight case. Both loose initial beliefs and perfect foresight lead to a diminished ability of the model to fit forecasting anomalies, accompanied by an even greater deterioration in its ability to fit the asset pricing moments.

In the first two cases, we center the 1996 median beliefs of software around nonsoftware beliefs but allow for wider distributions of initial beliefs. The first case uses a wide distribution that doesn't encompass the 2023 posterior median values, while the second case further widens the distribution to include these values. Both scenarios yield positive forecast error biases for software companies, similar in magnitude to those observed in the data, while nonsoftware forecasts remain unbiased. As we increase the support for the priors, the statistical significance decreases, with t -statistics dropping from above 2 in the

$g_{s,t}$, agents expect future revenue growth to be $\mathbb{E}[g_{s,t+h}|\mathcal{H}_{t-1}] = \hat{\mu}_{s,t} + \phi_s \rho^h x_t$, assuming for simplicity that ϕ_s , ρ , and x_t are known. After observing $g_{s,t}$, the cash flow surprise is $cf_{news,t} \equiv g_{s,t} - \mathbb{E}_t[g_{s,t}|\mathcal{H}_{t-1}] = (\mu_s - \hat{\mu}_{s,t}) + \varepsilon_{s,t}$, where $\varepsilon_{s,t} = \pi_s \sigma_{c,t} \eta_{c,t+1} + \sigma_{s,t} \varepsilon_{s,t+1}$. Note that the cash flow shocks are ex ante mean zero, since $\mathbb{E}[cf_{news,t}|\mathcal{H}_{t-1}] = 0$. However, if agents consistently underestimate the true long-run growth rate of software—that is, $\mu_s > \hat{\mu}_{s,t}$ —then the cash flow surprise $cf_{news,t}$ will be positively biased and predictable based on the information available at time t .

¹⁹For example, AI, which is another software offering, is initiating a new wave of technological innovation. Andy Jassy, CEO of Amazon, stated in Amazon's 2023 Annual Shareholder Letter that "Generative AI may be the largest technology transformation since the cloud (which itself is still in the early stages), and perhaps since the Internet."

benchmark case to around 1.70 and 1.50, respectively. Both cases successfully replicate other forecast anomalies, such as autocorrelation in forecast errors and Coibion-Gorodnichenko regression coefficients. Notably, the first case still produces model-implied alphas for software companies, albeit smaller in magnitude, whereas the very wide priors in the second case result in alphas close to zero.

The third case sets the initial beliefs in 1996 equal to the 2023 posterior median estimates, effectively endowing agents with perfect foresight. This scenario shows a significant reduction in model-implied anomalies for software companies. While biases remain positive, they are only marginally statistically significant, with t -statistics around 1.55. We still observe positive model-implied alphas for software companies, but their magnitude is also smaller relative to the data. The persistence of a positive alpha in the perfect foresight case indicates that anchoring the initial beliefs to the ex post posterior parameter estimates is insufficient to completely center them on the “right” location in the parameter space, implying that the true long-run software growth rates exceed those estimated using the whole sample.

The conclusion from these exercises is that for the Bayesian learning framework to quantitatively explain the persistent anomalies in forecasts and returns, agents must not have fully anticipated the extensive adoption of software technology by 2023. Instead, they had to learn about this trend gradually from noisy data. In this sense, relatively narrow priors are important in generating slow learning.

IV. Alternative Explanations

The previous section demonstrated that a Bayesian learning framework with priors miscentered around the mean of nonsoftware companies growth rates can account for the observed anomalies in forecasts and returns for software and nonsoftware companies. This section examines alternative explanations and discusses why they are less convincing in explaining our results.

A. FIRE Framework

The evidence that a single industry has outperformed beliefs for three decades challenges the notion of FIRE. FIRE assumes that agents fully understand and observe the models generating growth rates for both software and nonsoftware companies, including their underlying parameters, latent states, and realized shocks. Using this information, agents form expectations and generate optimal forecasts that should be correct on average, as they are based on the known laws governing company growth rates. This framework should therefore preclude systematic forecast anomalies.

For FIRE to explain our findings, it would require that these rational forecasts turned out to be incorrect due to realized shocks occurring between the time of forecast and the actual outcome. Specifically, it would mean that software companies experienced a streak of persistently positive shocks for 30 years, leading to their outperformance and resulting in positive biases in forecast errors and abnormal returns. Naturally, the likelihood of persistent, one-sided shocks over such a prolonged period is extremely low.

It is also challenging to explain our results related to the returns and alphas of software companies using standard rational macrofinance models with time-varying risk premia. In these models, the main pricing equation typically takes the form

$$(10) \quad \mathbb{E}_t[R_{t+1}] - R_f = -R_f \text{cov}_t[M_{t+1}, R_{t+1}] = f(S_t),$$

where M is the stochastic discount factor, usually a function of state variables S .²⁰ Under rational expectations, $\alpha = 0$ as expectations must, on average, equal realizations. A caveat to this analysis is whether we have correctly accounted for all appropriate risk factors when computing the alphas. This is relevant as these risk factors are not directly observable, and investments in new technologies can affect both firm growth and risk profiles (Babina et al. 2023). While accounting for additional relevant risk factors might reduce our alpha estimates, it would not explain the systematic biases and predictability of forecast errors documented in our empirical section, which remain a key component of our empirical analysis.

We now turn to prominent alternative explanations from the literature that deviate from FIRE and have successfully accounted for forecast anomalies in other contexts.

B. Informational Frictions

A common explanation for forecast anomalies is the presence of informational frictions. In sticky-information models (Mankiw and Reis 2002), information rigidities occur because agents only partially update their beliefs when new information arrives, due to the costs of accessing and processing it. However, these models, while arguably applicable to households, are less convincing for professional equity analysts as they have been trained to find and analyze the most current information. Equity analysts not only have the capacity to efficiently access and process information but also benefit from direct access to top executives. Furthermore, the sticky-information friction is even less plausible for C-suite executives, as they are the most informed about the company's financial performance, yet we also found large biases in these executives' own forecasts.

Information rigidities can also emerge in noisy-information models (Sims 2003; Woodford 2001; Maćkowiak and Wiederholt 2009). In these models, agents understand the structure of the model and the underlying parameter values, and they are rational at the individual level, updating their beliefs using Bayes' rule when observing noisy signals. However, forecast anomalies arise at the consensus level due to the gradual adjustment of all agents' beliefs to new information. As Bordalo et al. (2020) explain, maintaining rationality in individual updating implies that forecast anomalies should not be present at the individual level—forecast errors should be mean zero and unpredictable. This implication contrasts with our findings, where we document similar anomalies at both the consensus and individual levels (see Tables 2, 3, and 5).

²⁰For example, S is the surplus consumption ratio in Campbell and Cochrane (1999), the conditional volatility of expected consumption growth in Bansal and Yaron (2004), and the time-varying probability of a large drop in consumption in Gabaix (2012) and Wachter (2013).

C. Departures from Bayesian Updating

Relaxing the assumption of Bayesian updating can generate anomalies in forecasts at both individual and consensus levels. Bordalo et al. (2020) propose a model of diagnostic expectations where forecasters overweight states that have become more likely based on their noisy signals. With sufficiently noisy information, diagnostic agents discount their own signals, leading to rigid consensus forecasts. This pattern of individual overreaction and consensus underreaction relative to FIRE aligns with observed behavior for several macroeconomic variables.

However, this model cannot explain our findings. First, standard diagnostic expectations models typically exhibit unbiased forecast errors (as shown in Table 2 of Bordalo et al. 2020),²¹ whereas we document large positive biases for software growth forecasts. Second, these models predict individual overreaction, yet we find individual underreaction (Table 5, panel D). Bordalo et al. (2020) argue that Coibion-Gorodnichenko coefficients should be more negative for less persistent series, such as long-term versus short-term series. This pattern seems plausible, as Bouchaud et al. (2019) found individual underreaction for short-term forecasts, while Bordalo et al. (2019) documented overreaction for long-term forecasts. To test this, we present Coibion-Gorodnichenko coefficients for long-term (one to four years) forecasts in Table A.17 of the Supplemental Appendix. Consistent with our previous results, we find underreaction for long-term forecasts at both individual and consensus levels, with the estimated coefficient increasing with the forecast horizon.

Bordalo et al. (2020) builds on a large literature that deviates from Bayesian updating, arguing that forecast errors can result from inefficient use of available information (e.g., Mincer and Zarnowitz 1969; Friedman 1980; Nordhaus 1987; Maddala 1991; Croushore 1998; Schuh 2001). However, these alternative explanations are also conceptually less suited to explain the behavior of professional equity analysts and C-suite executives, who face substantial financial incentives to optimally utilize available information. Furthermore, it remains unclear how these theories of forecaster irrationality could provide a cohesive framework to simultaneously account for our evidence of large positive forecast errors for software companies and unbiased estimates for nonsoftware companies.

Introducing behavioral biases could improve the model's predictions for short-horizon forecast errors. For example, Li, Van Nieuwerburgh, and Renxuan (2023) find that diagnostic expectations or overconfidence can remedy these shortcomings without interfering with the longer-run forecasts, which are driven by learning.

D. Incentives of Analysts and Executives

The explanations discussed thus far assume that forecasters aim to minimize the mean squared error of their predictions. However, analysts may have incentives to bias their forecasts, meaning that their forecasts might not be equivalent

²¹To understand this, consider the forecast error in Bordalo et al. (2020): $s_t^i - x_{t|t-1}^i = s_t^i - x_{t|t-1}^i + (1 + \theta) \frac{\Sigma}{\Sigma + \sigma_\epsilon^2} (s_t^i - x_{t|t-1}^i)$, where θ represents the degree of departure from rational updating, and $x_{t|t-1}^i$ is the rational forecast of the hidden state as determined by the Kalman filter. Since rational news $s_t^i - x_{t|t-1}^i$ has zero mean, forecasts display no average bias (i.e., $\mathbb{E}_t[s_t^i - x_{t|t-1}^i] = 0$).

to conditional expectations. We now examine whether the various incentives for forecast bias proposed in the literature can account for our findings.

Analysts may have an asymmetric loss function (Capistrán and Timmermann 2009), where the costs of over- and underpredicting a variable differ significantly. This is particularly relevant in our context, as analysts may bias their forecasts to curry favor with management for better access to private information (Ke and Yu 2006). Additionally, reputational concerns may lead analysts to engage in forecast smoothing, herding toward consensus, or intentionally deviating from consensus to stand out. These incentives are not limited to analysts; C-suite executives may also have reasons to bias their forecasts. For instance, they might issue conservative guidance (Bartov, Givoly, and Hayn 2002; Skinner and Sloan 2002; Matsumoto 2002), allowing them to “beat and raise” analysts’ forecasts and avoid stock price declines associated with missing growth expectations at earnings announcements.

Although forecasters’ true objectives are difficult to observe, our results challenge explanations based on reputational or other distorted incentives. Such incentives would need to (i) differ between software and nonsoftware companies to reconcile the positive biases in forecast errors for software companies and the negative or unbiased forecasts for nonsoftware companies (see Tables 1, 2, and 3); (ii) vary at the individual analyst level to explain the systematically positive forecast biases for software companies but not for other nonsoftware technology companies followed by the same analysts (see Table A.9 in the Supplemental Appendix); and (iii) change over time to explain the learning-like pattern presented in Figure 3. Furthermore, these theories would need to explain why investors have not learned about and unraveled these biases, given their persistence over several decades and our documented abnormal returns that align with the forecast errors.

V. Conclusion

This paper examines the valuation of software, which has emerged as one of the most disruptive and important technologies in the digital age. We document that companies primarily valued for their software offerings show a disproportionate increase in market share, growing from 2 percent of US equity markets in 1996 to 13 percent in 2023. During this period, we uncover persistent anomalies in growth forecasts and stock returns for software companies. These anomalies indicate significant deviations from rational expectations over nearly three decades.

We find that analysts’ individual and consensus forecasts consistently underestimate the revenue growth rates of software companies by approximately 2 percent per quarter. This systematic underestimation is not limited to external analysts; similar forecast errors are observed in growth estimates issued by company management. Moreover, we find that stock prices of software firms exhibit positive drifts that align with these forecast errors, generating buy-and-hold annualized alphas exceeding 400 basis points. We do not observe similar anomalies in other companies.

We show that our results are consistent with the Bayesian learning literature, which reconciles anomalies in forecasts and returns with rational updating. In our framework, investors gradually learn about software’s growing importance, leading to persistent anomalies over many decades. Agents in 1996 did not foresee the large-scale adoption of software in the subsequent three decades; instead, they

learned about it from noisy data. We also examine alternative explanations based on informational frictions, departures from Bayesian updating, and incentives of analysts and executives. We conclude that these alternative explanations cannot fully account for our findings.

The study's findings not only reinforce the challenges in measuring and valuing innovative assets and intangibles in the new digital economy, but also illustrate that markets can be very slow at discerning structural shifts from transient shocks in noisy data, even after several decades of learning. The occurrence of anomalies in these learning frameworks extends beyond technological revolutions and is closely tied to specific characteristics of underlying data. Such episodes are likely when persistent variables undergo gradual shifts over extended periods, ultimately resulting in substantial changes that diverge from historical patterns.

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